

Krys Dolega ~ Active Inference Insights 022 ~ Emperor's New Markov Blankets, Attention, Dan Dennett

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hello everyone and welcome back to the show today I'm extremely pleased to be speaking with Chris deler Chris is a post-doctoral fellow working in the center for research in cognition and neurosciences in Brussels he focuses on computational models of cognitive functions mental representations and Consciousness as well as broader topics prominent in philosophy of computation philosophy and ethics of AI metacognition and philosophy of science Chris thank you ever so much for joining me um thanks for the invitation oh it's an absolute pleasure um people have been asking for a bit more of uh you know critical voices on active inference which uh and and you're the perfect person to go to but before we get there uh we were talking a little bit off air about Dan Dennett um Dan passed away unfortunately last week at the time of this recording I believe it was and he was your PhD supervisor so clearly a great deal of your work has been influenced by Dan um I wanted to talk about your 2015 paper um discussing the cartisian theater and it's sort of place or lack of you know it's it's sort of ought to lack of place in predictive processing or perhaps not but before we go there um I know you want to say a couple of things about Dan and it would be great just to discuss his sort of influence on you more more broadly sure thank you so yeah so this is a really sad moment because that has I think it's easy to say defined in large in many ways defined the the shape of philosophy of Mind in the 20th century up to today and it's a really great loss to to see him go um and I I guess I I mentioned to you when we were setting up this meeting that I would like to spend a little bit of time discussing him because uh I think that Dan is in some ways uh a kind of a philosopher who should be uh followed in the way that he approached philosophy of mind and uh I think that it's you know worth focusing on that that's that's one thing that I find really interesting and important about him and the other thing is that uh I had a personal relationship with him uh I did two extended research stays with him during my PhD and we've met at many occasions at different conferences uh and uh over the past weekend

there were a lot of obituaries uh that that showed up about Dan and uh he clearly was a lot of things to many people and it's actually astonishing to see how many people he has influenced across multiple disciplines but I think it's also worth remembering uh that he was just an extremely kind uh nice guy he was just a he was just a joy to to be around um and so that's the second thing that I sometimes feel is missing from contemporary philosophy so so the way he did philosophy is that he tended to focus on science right and he thought that there's no philosophy free science uh and similarly he thought that that philosophy of Mind should be informed by uh our best state of scientific knowledge and sometimes I feel philosophers forget that it's very easy to uh stay within the boundaries of our discipline and and not reach out uh and not try to to you know have a peak what's going outside of the boundaries of the Ivory Tower of philosophy of mind and I think it's really important to remember that that it that you know that if there's a legacy to Dan's work it's that philosophy can be informed by science and it's actually very productive and important for philosophers to engage with scientists and it's a kind of a two-way it's a two-way uh exchange of ideas where both sides can benefit productively um and then the second thing is that philosophers can be extremely serious about themselves and about their work and about really entrenched in their positions and and uh very competitive uh and uh yeah kind of hardheaded and I think that Dan uh although we cannot say that he had a small personality he was definitely uh you know a flamboyant personality as well but he he was really a person who was very kind towards Junior Scholars he was he would always find time to engage with with uh younger researchers and students he was very curious of what other people might say about his work even if it was critical and uh he did not take himself entirely seriously and I think this is something that's also worth remembering right so he he was happy to sometimes concede that maybe he didn't think something through uh and uh he thought that first philosophy first and foremost should be a great intellectual adventure and not some kind of dogmatic holy war and I think that's that's something worth remembering I don't know if you would like to ask anything else about Dan or well I um yeah I mean firstly to Echo everything you just said I mean I never met Dan um I guess I'm I'm not currently at that place in my career where that would have been viable I mean um I only left school what you know uh in 2017 but you know even studying philosophy at school Dan's name was pervasive right I mean interestingly pervasive across sure philosophy of mind which we studied but even qu you know broader questions of epistemology um an ontology Dan's name would come up in a way that other philosophers wouldn't and so he always seemed like a kind of Grandfather of modern philosophy in you know and so he's one of for me one of the sort of

icons who I've never met and I've never spoken with but are incredibly influential even if as you say not I don't agree necessarily with everything I think Dan thinks or thought but yeah incredibly important um and you're absolutely right I mean just being on Twitter or X or whatever this weekend it was unbelievable I I mean maybe in some ways this is just because I follow so many academics but just the timeline was absolutely replete with with uh obituaries and condolences and it was quite remarkable so clearly he touched a lot of people um both as a human being and also as an intellect yeah I'm curious what was he what was he like sort of during your PhD uh I mean I guess this is a personal question because I that's what I want to be embarking on you know this year but what was he sort of like as a as a guide and a mentor so he was extremely uh kind as I said before so so uh very encouraging he was eager to learn so something that I've heard from other people who have done post appointments with him or have been his PhD students in the past was that when someone came to him with an next with a expertise in a field that he wasn't very familiar with he was really eager to learn uh from them so so when I was working on my dissertation which was on predictive processing um and that was a new topic back then um he was very interested in learning about the models and And discussing with me formal papers right about about different modeling algorithms for example which was super nice yeah um and one thing that was really special about him as a as an academic Mentor was that he was uh extremely well-rounded so I feel that today in Academia I mean it's not a product of the last 10 years but you know we we we've we have very narrow focuses we we we value specialization and then explored everything so a lot of people don't remember this about him but he was able to code for example he has done work with programming in the past uh he was very skilled in many domains and that would often come back because his curiosity and his abilities in in multiple areas would allow him to draw connections between different fields of work so so he was very quick in suggesting readings for example that you wouldn't get as standard recommendations from people working on a particular topic because they would be off they would often seem like they are very removed from what you're interested in but later on you would follow up and you would find out that there are actually connections between these ideas and that certain ideas surfaced in many different places in many different disciplines in Academia over time in different guises so that was extremely valuable uh and yeah he was just uh he was just fun to be around so he was obviously very busy he was still quite generous with his time despite how many appointments he had and how many people were always kind of you know fighting for his attention and trying to to to get a bit of his time uh but one of the best pieces of advice I ever got from him

was when I was visiting him and I was writing on writing a paper I was working on a paper and as I was halfway through this paper I don't know three quarters in someone else published a paper that was extremely similar to my paper to the point where I kind of thought okay I have to abandon this project because there's no way to publish this right now because it's just too similar and I was pissed I was irritated I was kind of you know I can imagine I sunk a lot of time into that project and I went to him and I said yeah I'm I'm pissed this this just happened what am I supposed to do uh and he said to me something that I try to remember every time you know I feel scooped by someone with a paper or or um someone is publishing like crazy on some topic that I'm interested in and I don't have I don't feel like I can catch up sometimes uh with the literature which has happened with predictive processing quite a bit my PhD uh because that literature exploded at the time but he told me this he said philosophers arguing about who came up with an idea first are like sailors wabbling over who was the first one to notice that the wind has changed and I thought that's that's very good advice because the reason we all got into philosophy wasn't probably because you know we wanted to be uh a superstar of philosophy it isn't because we hopefully it isn't because we have humongous Egos and we think that we are right and everyone else is wrong but it's just because we appreciated thinking about certain problems and it's in at the end of the day it's a collaborative Endeavor and uh you know science philosophy uh building knowledge is is a project where which in which we should engage collaboratively and despite being a giant that he was he was also then I mean he was always very uh uh in his writings and in in talking to him he was always very much against the idea of a singular genius so this this idea which philosophy often suffers from even today this myth of a you know single author that can lift a whole field it's a bit ironic that Dan was the person who was against this idea given how much influence he had over the field but he and this is visible in several of his books and writings he didn't like this idea a singular genius he thought that you know really certain ideas uh start to kind of Brew in in in the right conditions and then there's just someone who who who makes the discovery or who catches wind of what's happening but he always thought of himself as standing on the shoulders of giants like Quin uh and and and other people so yeah yeah that's a beautiful question qu it makes me think there's a there's one by KL young where he says people don't have ideas ideas have people yeah Yung was thinking about in a slightly different way than Dan was but it's um you know in in some psychoanalytic manner but the the sort of sentiment is the same yes and it's actually some it's extremely close to some of the ideas that he presented in from bacteria to B and back so so when he talks in the chapters where he talks

about memes yeah it sometimes goes extremely close to to that line of thinking that uh we I was doing my first research stay when he was finishing that book and I had a privilege of of uh going through the drafts And discussing the drafts with him and Martin bowry who was a post at the time with him also a visiting post do and uh I remember that we s one evening and we had this conversation where we picked a passage from the book from the chapter about memes and we said Dan it's what are you sure this is this is what you want to say here because it starts to sound like humans are just you know material vehicles for memes and is this it sounds very extreme and I remember he was sitting there quietly for a second thinking and he said you know what maybe you're right maybe my Pros got the better of me so um yeah so so to say right people always uh praised him for his Pros right um famously are kind of very well praic is a derogatory word but I mean it here in a in a sort of praiseworthy way that you some philosophy is very snooze worthy let's be honest uh but you know the things that I read of Dan are certainly not and that itself is quite remarkable um you know the ideas aside of course which are very potent and influential to be able to write about something otherwise deeply complex and otherwise for you know a certain audience extremely boring and to make that come alive is an incredible skill in of itself I yeah go ahead I just wanted to say on that point that I completely agree with you and uh he definitely had his own voice as a writer which was remarkably close to the way he would speak in conversations uh but what's really interesting is that that way of writing that he introduced into philosophy uh it was it it it gained praise over time I think late in especially in his later years but initially it was a bit of a problem for him so when he wrote his first book content and Consciousness which I believe came out in 1969 that book got some negative reviews because of the way in which it was written and it wasn't a fully actualized denial narrative yet uh but you could still get some you know of his uh his kind of Massachusetts sensibilities in there and and what was really interesting is that uh some people really uh recoiled from that style of writing at the time well I think um yes I I sense there's been a change in that right I mean the the underlying soci sociological reasons for why that's the case whether it is just uh people copying someone like Dan is an open question but I do think that um well Duty might be a strong word but we have a we we do have an obligation in some sense to make our works leg right uh even if it's not for the L audience even if it's being published in philosophy journals we can't be talking into you know this the N of one um you know V chamber that is just our own mind and Dan again of the stuff I read from and also his contemporaries uh people like Ned L and and David Charmers and then some of my favorites you know Evan Thompson who I know was a postto with Dan it's it's fun to read um

and it makes people like me want to do philosophy and again this is no shot at anyone who was working 120 years ago whose Pros might make you want to fall asleep but um I yeah I guess there's a broader broader thing there that um we do have a we do have a duty to transmit our ideas in a legible intelligible Manner and Dan epitomize that definitely and he was also I mean he was also a he had the style of his own which was very entertaining which I think is another thing yeah a lot of a lot of academics could learn from him uh he's not the only example of that um but but of of of you know of being able to write in an entertaining way while also transmitting very complex ideas but uh yeah I like this idea of an author having their own style you know um I think Carl is one of them I mean it's fitting that we on this podcast right I mean um I've been accused at certain points in my writing of writing in Frisian E I love I love review's comments where they mention Frist e because I can't help you know having read so much of Frist's work of Carl's work I can't help but use the word evince yeah that's definitely that's I feel like it's a bit of a dog whistle you know it's a if if someone uses the word evince in their paper especially when you're reviewing a paper you can see who they've been reading uh a lot so so I feel like evince is this kind of Mark of of someone who has read a lot of Friston if EV appears in your own writings it might be because you've been reading a lot of Friston and moving away from Dan a little bit and and towards what you wanted to talk about in terms of papers I reread some of my older papers for this meeting because you know some of them are seven years old and I don't remember what I wrote there uh I was yeah definitely it's a journey for me as a non-native speaker to develop some kind of style and and I I read some of the sentences I wrote in the past and I wouldn't put those ideas in that way today but you know well I've got to say that I didn't have that impression um the one I read I think it was yesterday was the The Curtain Call for the Cartesian theater which was 2015 which feels I mean like yeah I was doing my GCS for people in England you know I was doing my sort of I was 16 or whatever but um I didn't feel that way at all and and clearly there's a lot of debt and influence there in in that paper but what I like about it I was going to say this in reference to Dan is you know you mentioned his kind of openness his curiosity his epistemic humility fundamentally you know I mean we don't know with with a lot of things in life we really don't know and that's the beauty I mean we would all be out of jobs if we knew and life itself you know there would be no information transfer arguably there would be no life at least worth living um and I felt that in that paper you very much evinced if you pardon me the word a great deal of epistemic humility which I really liked I mean it you know you're critiquing the Frisian in this case the Frisian Cartesian model of Hobson model of um their their sort of

invocation of the cartesian theater and what Dan called cartesian materialism when it comes to I believe you call it action orientated predictive processing but I guess we call it active inference now yes but you know you didn't say but this is the this is the fundamental way in which Consciousness appears or this is the way the the representations work so I enjoyed that I mean for people who haven't read that paper what and again it comes back to Dan what is cartesian materialism um what kind of role does it have to play in terms of predictive processing and then what's the what's the problem with it you know okay uh nice so we're really starting at the beginning so that's a paper uh me and my good friend Joe with whom I studied in edinburgh before starting my PhD uh we studied in edinburgh uh where we read a draft of Andy Clark's surfing uncertainty and one of our Master seminars wow and that's how we got into processing um so so that's a paper we wrote early on and it was a kind of a knee-jerk reaction to a paper by Carl friston and uh Hopson yeah H in which they invoked the notion of cartesian theater so cartesian materialism is this term that was coined by Dan denet to describe a certain view of Consciousness which he thought was very a kind of a residue from the from the ideas of daycard and you know the idea that there's a certain point in the brain where information needs to reach and then it somehow needs to be rendered in the medium of Consciousness so for for decart that was of course the pineal gland which was this point of interaction between a mechanistic mind uh and the soul and there was a there was a certain in his view of the mind and his of the brain also uh there were these uh humors right there were these uh different liquids in the brain that would create pressures on the on the pineal gland and depending on which part of the pineal gland they would put pressure on there would be a certain experience that would uh appear in in in the soul in in Consciousness uh so so what denet criticized about this view of Consciousness is that it requires a certain type of double trans transduction he called it so it's not only that there's sense it's not only that there are forces impinging on our sensory periphery from the environment which are then transduced into the medium of neuronal spikes and you know um hormone secretions and so on all the things that are happening mechanistically in the brain but then it seems to imply that there's a certain point in which in the processing of information where the medium information needs to be transduced again from the medium of neuronal spikes and all the mechanistic um happenings in the brain into the medium of Consciousness so he sometimes referred this this this the product or the this this this substance of the second transaction he called it figment right with with by analogy with pigment so figment is this thing that needs to be put on to your you know onto your representations of the world in order to imbue them

with qualitative properties because presumably as most dualists and non-reductive physicalists would tell us there are phenomenal properties and there are dispositional or functional properties and these are two different types of properties right so all anti-reductionists about Consciousness will claim that phenomenal properties are not reducible to functional properties and then said well if that's the case then it seems to be then we end up with this picture of the Mind where information hits the retina let's say in the case of vision it's transduced into the medium of neuronal spikes that information is those neuronal spikes are propagated into through the cortical hierarchies and then we have to ask where is this point in the brain where we put on the pigment onto the visual information right so so it seems like we need to apply the mental paint as block would put it again even though I would like to say that here as I'm talking about mental paint that's a slight abuse of the term as block uses it but the point is that you need to you seem to require adding something into the workings of the brain in order to render information consciously information conscious and that's something that he really wanted to reject so his point uh wasn't that there's a his point wasn't that there's there cannot be some aspect of the brain which is privileged in terms of bringing about conscious experience H it's rather that there's nothing extra in ter metaphysically extra or ontologically extra added into the information when it becomes conscious right it's just that you have you process information unconsciously and then it can become conscious at any particular time uh given that it was processed enough through the cical hierarchy and so on right so so there's no mysterious properties that need to be applied again or added back into the into the information that is being processed by the physical system so he very much so the name cartisian Mater materialism is supposed to suggest that there's a there there's a materialist picture of the brain but there's this much like in the case of the pineal gland in in the case of thearts dualism in cartisian materialism there's a particular location in the brain that information needs to reach in order to become conscious and this and this particular Center in the brain is this one area that serves as the cartisian theater kind of as a as a place where which serves as the screen of Consciousness right and that's something that he wanted to reject because he believed that there's no one particular place in the brain where information needs to be processed in order for it to be conscious it's rather that there's a collection of different functions that need to be in place simultaneously trading information in order for information to be conscious right and that's also why in many disorders of Consciousness such as I don't know visual form agnosia uh or blindsight we see that conscious experience underg goes a kind of graceful degradation where people lose certain aspects of

conscious experience uh but that doesn't mean that they are rendered somehow entirely unconscious or something like that right it's it's rather that you can knock out different modules if you want to talk about that in this those ways are different systems out but as long as there's some kind of wider Comm of information going on throughout the brain Consciousness is still in place so I feel like the rebuttal to this would be the argument that okay I guess the Cartesian theater in some ways has multiple reference in the sense that one thinks of as you say the screen of Consciousness I mean the word theater brings that to mind very vividly uh and one thinks also of the homunculus who's watching the the screen and this sort of little man who's watching and and that's obviously got a number of problems most strikingly the infinite regress of the homunculus but then one but then there's also this separate more mechanistic point about the locus of Consciousness and this sort of singular physical point I guess the rebuttal to that would be well even if you adopt a more distributed idea of the brain as a network and you're talking more in terms of functional connections there still seems to be either you know either we're saying well there is no consciousness well let's disparage with that just for the meantime but let's say there is consciousness well we don't want to say that necessarily the sound waves or the visual information itself is conscious so there does seem to be a transduction at some point from matter into phenomenal space now that sounds like to me that there needs to be a moment at which just by necessity that information enters into you know in metaphorical use that phenomenal space now it might not be the case that that's the same for every experience it might not be the same that it's always it might not be that it's the case it's the same in every place for you know for every experience but it does seem to be this case that even in a diverse range of networks there is a moment at which the lights come on so to speak would Dan or yourself have a problem with that weaker form of the Cartesian materialism or Cartesian theater um all right so that's a good question that's you're pinpointing exactly the the the main issue so so um Dan I think would have a problem with that because especially so so he explicitly wanted to uh leave behind this metaphor of a screen mhm or this metaphor of uh you know know some kind of special properties in in Consciousness so a lot of people attack him these days for being an eliminativist about Consciousness and he was very adamant that he's not in that camp precisely because he would always claim you know there is something it is like to be Dan Dennett and it's different from what it's like to be me and it's different from what it's like to be you so there is a subjective perspective on the world that you have that is that is different from mine and that was different from his uh and uh so he didn't want to deny that but he wanted to change the way we speak

about these things so so this is why he together with Kei Frankish they coined the term illusionism because they wanted to stop talking about phenomenal properties as things that we have to uh find a place for in our ontology now uh yeah so that's that's one thing and I think that he would also have a bit of a problem with that because uh of his multiple drafts model so so his idea was that you know he was very much informed by uh bars and thean and the GL the work on global workspace so this idea that there's this there is this there are these longterm long range projections sorry long range projections in the braak which connect different areas and there's a kind of exchange of information going on but on his view of Consciousness there isn't one particular system that is the Consciousness system right so so it's it's likely that I think he would agree with the claim that a weak claim that there is a number of fun cognitive functions that need to be in place in order to for something to have Consciousness like ours or sufficiently similar to hours um but I think he would reject the uh the claim that any one of these systems is somehow privileged over the others that's that's one thing and then the multiple drafts idea is that there are multiple competing streams of information in all of these systems that are necessary for Consciousness and which stream you're conscious of which stream of uh you're experiencing is kind of up to one the the the how much Cloud it has in the system so how strong how much control it has over over the rest of the of the cognitive economy what he kinded as Fame in the brain and then the other thing is the kind of probes that you encounter so so kind of uh either external um external situations which which or stimulite that that that require solicit a certain response from you or internal probes like you know if you're if you're sitting there and you're really thinking about your uh ex or if you're really thinking about the EXP focusing trying to be mindful about the experience of eating an apple that's also a way of privileging certain streams of sensory information or or memory traces over other streams of information in the in the brain um and so okay so that's I think what he would want to say probably about this now for me I guess I'm less uh I'm less uh convinced of of you know I I like the multiple drafts model but I'm not completely on board uh because a lot of the work in science of Consciousness these days focuses on uh on higher order systems on this division between lower order systems and higher or lower order processing and higher order processing metacognition um and I think that I'm leaning more towards some kind of higher order theory of Mind these days and I do think that there is a there might be a certain privileged set of processes or systems that are are necessary and in some sense they might that they're not exactly I wouldn't want to slide back into cartisian materialism by saying like oh information needs to go there in order to be

conscious but they do play an important functional role um and I think that because I I I subscribe to some version of the higher order View I think that means that I have to be a little bit weaker on the cartesian materialist critique this these days because there is a certain I do separate the system into two levels and where one level is seems to be privileged over the other in terms of uh the role it plays in bringing about Consciousness okay cool we can go there that's that's that's an interesting development because um I'm so I'm aware of these higher order theories of Consciousness with respect to self-awareness so as you know there are these so-called uh pre-reflective states of um self-awareness or self a subject which you can sort of deflate just a subjectivity let's say and the whole the whole principle of these you know phenomenologically and ontologically is that no one there is nothing that is aware of them they offer the uh texture the background to uh the objects of Consciousness but they never areed into an intentional object obviously his work goes back to S ponti H and so on now uh this kind of poses an issue for the higher order theories of Consciousness because for them these states or or or these experiences which in turn are not reflected upon they well it seems a bit problematic because they propose that actually there are higher order states which take these conscious uh these pre-reflective states as their object so there is no such thing as the pre- I've or non-reflective self-awareness rather it's sort of hidden hidden uh these higher order uh intentional as are hidden from us and then the rebuttal comes and says well how you know how are these not conscious right like surely I should be aware of these higher order states which have as their object my pre-reflective states of self-awareness and the higher order Consciousness theorist come along and say Well they're not conscious uh and they intend the pre-re states of self awareness without themselves being conscious and then you have just the ontological question well how does that work and so I guess the question would be we seem to have these it doesn't have to be self-awareness we seem to have these uh phenomenal I guess the word coming to mind again is textures right I mean what people traditionally would have called qual or Quail which is sort of um what you know seems to constitute the what it is likeness of Consciousness so the redness of red or the APPL of eating an apple and so on and um these seem to be constitutive of my experience even if I don't necessarily have to rify them and that's the same with sort of the my own subjectivity or my own intentionality how do you as a kind of leaning towards a higher order Theory Of Consciousness how do you deal with these otherwise seemingly irreducible ineffable intrinsic features which at least on the surface seems to seem to Bell um becoming an intentional object a sort of overt reflective object uh all right wow uh that I'm no no no no no that's that's that's a good question um so I

must say that I'm not that familiar with the literature on the pre-reflective uh self-consciousness I'm I'm I'm getting more into it because I was recently interested my my recent work is more interested in affect and effective experiences so that's what I work on in Brussels now and some people postulate that for example in order to have emotional experiences you need to have a an explicit involvement of the self yeah so jolu for example is a higher order theorist who claims that in order to undergo an emotional experience you not only need to activate what he calls an emotional schema uh which conceptualizes your the the sub uh subcortical processing in the amigaland and so on but also you need to have an active self schema so you kind of his big big catchphrases no self no emotion so unless you you are aware that it's to you that something is scary then you just don't feel fear right previously he call like the smaller uh claim was no self no fear so you can only fear something if you realize that is dangerous to you or or you you fear um that thing that object so so I'm not yet that familiar with that literature and I know also that even if we leave the self aside there's this whole literature on minimal phenomenal experiences which is something that Thomas metzinger is now very interested in um precisely because in deep meditative State you can let go of the of the self right and and you can presumably have these pure phenomenal experiences uh which are supposedly uh relaxing this idea that there's a self which is the the anchor for the point of view on the world um and I think that you're right that both uh minimal pre-reflective uh experiences uh this this and and minimal phenomenal States they they do put pressure on on higher order theorists and so I think that there's two ways in which we can respond so so one way is I think the way that would be taken by someone like David rosenal so David rosenal talks about qualitative spaces and he very much wants to uh wants to uh reclaim the word qualitative property from the hands of of strong phenomenal realists and say look there's nothing mysterious about qualitative properties as such qualitative properties as such are not uh some kind of mysterious properties they're just we know about qualitative properties we talk about qualitative properties in science and we can think of the brain as representing qualitative properties that's not the problem regular scientifically understood qualitative properties are not problematic on their own in in in a deeper metaphysical way unless you conflate them with qualia which are supposed to be not only qualitative but also you know they're supposed to be ineffable they're supposed to be private and so on intrinsic and so on and so forth so what he's trying trying to do is he's trying to separate the notion of qualit qualitative property from the notion of intrinsic property and once you do that you can actually claim that on the first level you will have representations of qualitative properties and

then you need this kind of higher order process that targets or picks out or re-represents depending who you are these lower order States in order to have qualitative experiences and that's that's one way you can go about it I agree that this might not be satisfying to you because the problem then is well okay so what about these states where we seemingly undergo ego uh dissolution and and where the self seemingly doesn't play any role it seems to me that in those cases traditional higher order theorists will have to claim that well you're kind of in a under an illusion that there's no that there's no self perhaps The Narrative self Falls away but there might be this kind of minimal notion of a self that's still in place for example presumably if you're in a deep meditative State uh you're still deploying your attention in some way or controlling your attention in some way there's some kind of process of attentional monitoring happening and the system which is controlling the the the allocation or the flow of attention if you want to call it that that's the system which is actually responsible for for this minimal self that's where the minimal self resides and and Michael graciano as you probably know has built a whole view of Consciousness around the idea that the control model for attention is the the the the representation that we introspect when we're thinking when we're experiencing things and then the second way so that's one way to do it the other way would be to say well how many higher order levels are there right so you can always this is a you know it's as much of an objection as a as an argument in favor right so you can always claim that there's an extra level of representation and it's not that things have to be represented on the second level there's a higher level and so for example in graciano's model to go back to that as I understand his picture these days we have have some first order perceptual or or other processes uh then we have the attentional model which we represents these systems in a kind of uh way where they are distorted uh and then we when we introspect there's actually a higher order model yet that is picking out the contents of the attentional control model uh at least that's one way to understand his claim and so in that picture we would have this kind of of three layers uh of of the higher order model right so it could then we would we could claim for example that when we're having these pure minimal experiences that the higher the higher level that this introspective level which which renders things as for me experiences is dropping out but the control model again is still there in place and so what we're experiencing is just these representations in the control model but there's no explicit level of introspection so the self is kind of sidelined I'm kind of spitballing here right now no so so was I to be fair because as I said I'm not uh first of all I don't have a complete Theory Of Consciousness and I would be I hope the last person to claim that maybe one day uh you know I will feel more confident in

claiming some piece of the cake for me for myself sure um but but and I as I said I'm interested in these minimal States uh of either pre-reflective self-awareness or minimal states of phenomenal experience but I don't know enough about them right now to you know give you a very um strong position about them yeah I mean to be thought I was somewhat spitballing as well I am not an expert in higher all the theories of Consciousness I mean I'm aware of David Rose and others who have proposed things akin to a high World theories of Consciousness right I don't really believe in sort of Monolithic these kind of categories right everything is sort of everyone's got their own idiosyncratic way of viewing these things um I I came across it I I'm very I am interested in pre-reflective self-a Awareness kind of phenomenologically more than anything I mean the ontological metaphysical ground of it is an interesting question I I mean just to State my claim at least on something like subjectivity I think there's a it's absolutely viable to take a kind of more San or moonan position that it's actually just imminent it's intrinsic in all experience even in minimal phenomenal experiences there's still the for meanus and then I mean this feeds into my kind of skepticism over higher order theories because even in this notion of a kind of higher order representation there is already inbuilt into that the for meus that is supposed to be being represented which seems to be kind of begging the question or you know putting the cart before the horse and I would again I mean this idea that it's just intrinsic to experience and then in some ways you're saying well experience is intrinsic fundamentally and there is just this irreducibility of course that comes with its own problems right I mean I mean how first and foremost I mean unless you're just a Jew list out the you know out the can like so so no these are interesting metaphysical questions I guess um it was probably it was probably a bit misguided of me to bring in the notion of qualia because well I guess this is an interesting question when I was thinking about it I was thinking about the way it sort of pre-reflectively uh structures our experience such that I I mean this again I don't really know if this is true do I need to reflect on the taste of coffee to experience the taste of coffee in the same sense do I need to reflect on my I don't well seemingly phenomenologically I don't need to reflect on myself as an object to experience myself as a subject um and so that's kind of why I bought in qualia but I think probably the higher order theories a better chance of dealing with qualia right it's just a representat of some represent you know representing scheme um yes I mean these are Dee rooted philosophical problems that we're not going to be able to solve on this podcast but I think an interesting question here is um bringing it into a basian account so you bought you mentioned s of Fame in the brain there uh another one of your papers s Fame in the basian brain yeah the question I

think that comes up just to mind is if we took multiple draft models seriously and it seems it's well in some it seems incredibly convergent with predictive processing and active inference in the sense that we're running multiple hypotheses of the underlying cause of our sensory data the thing is is that we in principle you're never reaching the kind of infinite peak of a gaussian you're never coming up to a you know no hypothesis in principle statistically should occupy a one from 100% certainty because this is just statistical distributions right yeah if we're taking a very Bare Bones approach to basian the basian brain hypothesis so I mean I've been reading this in terms of people like Andy Clark and Kate Nave have spoken about this kind of seeming in the fact that perceptual experience seems determinate or be it seemingly if we take the basian brain hypothesis seriously underpinned by indetermined probabilistic processes so I'm curious about how one can skirt this problem I mean Kate has her own solution which is that Consciousness in some sense is indeterminate although not in the way that people naturally think of indeterminate she thinks as a sort of singular reality nonetheless um people like Jakob Hy will talk about distrusting the hypothesis and so there's this constant temporal flow I'm curious about how you can get how you think we can get from a predicted processing picture of multiple hypotheses being entertained to a singular seemingly singular coherent global of conscious perception yeah that's a that's again a very good question um so I think I would start this by saying that I don't so so I'm very much interested in the basian brain hypothesis as a from a kind of anti-realist stance right so my stance towards towards these models is kind of um disengaged in a certain sense ontologically so I suspend my belief in them as actual descriptions of what's happening in our heads yeah and I think that uh I take this removed position of all models are wrong some models are useful some models are interesting uh and I think that actually probabilistic models have a lot going for them but it's just one of the possible ways of capturing what's going on in our heads uh and so we always have to so if you approach them from this this point of view then then you know first of all it relaxes a little bit your commitment to finding the one true solution so so you can say oh we could model this in this way or we could model it in that way we we make some modeling choices that in the end determine how we're going to think about it or how we're going to model the behavior of the system uh and in some ways they are useful but in some ways they can also be misleading right so um so that's that's that's the kind of to preface my my answer here so that's the first thing uh following from that one of the big decisions that you have to make when you're deciding how to model the basian brain If you assume that the brain is you know doing something like bayesian inference then of course the question is how are you going

to model the process by which it's doing it so is it we know that it's highly unlikely that it's running exact Vision inference uh so it needs to use some kind of method of approximation so Carl and people in his camp have settled on variational bayesian inference as the most likely solution here but there are other options as well right so you could think about sampling for example which is another way yeah and if you had a model based on sampling you wouldn't actually have to be like that the problem of determinacy is gone in many ways because if you think that you're you know sampling from some kind of distribution then you're and then you're experiencing a sample then you're experiencing a sample it's a determinate point and and kind of you the information is determinant right even though it comes from a probabilistic distribution so that's one way you could get around this um the other thing that I wanted to point out is that in uh yacob's picture of how things work so yakob talks about probability density functions of course um and uh he talks about some kind of uh as you said uncertainty about you know the exact uh hypothesis right that you're entertaining because you're never getting a kind of a pointwise estimate you're always getting some kind of interval um and that's that's also a way to go get out of it another way to get out of it is to for example bring a bring into the picture some kind of uh dual coding hypothesis so you could think that on certain levels of the predictive hierarchy things are being coded in continuous spaces and in others they are discretized right so some levels of the hierarchy are operating on uh on probability density functions and there are parts of the hierarchy that are operating on probability Mass functions okay where you can actually pick out one hypothesis as the as the currently winning hypothesis so the way you the reason why you might think that's the case is because of how Concepts have been typically handled in philosophy right so concepts are supposed to be these kind of distinct categories at least traditionally nowadays A lot has changed in the way we think about Concepts but you could think that as you're bringing something into a conceptual frame that conceptual frame is somehow constraining your experience so you could think that well when I'm perceiving colors of course the space of my color perception is continuous but then as I'm recognizing colors as I'm applying certain Concepts to them I'm discretizing my experience in certain ways uh the problem of course with that is that there seems to be a lot more variety in our perceptual experience than our conceptual appar this allows for so we both agree on what red is but we could find multiple examples of objects that we would categorize as you know we would disagree about whether they should be categorized as red or not right and there seems to be a lot of perceptual experiences that fall under the concept of redness for both of us and we would agree that there

is variability in the in what we call in the experiences we would uh uh label with the term red so so that's then a problem and I guess another option sorry I'm kind of lay layering you know different different ways you could think about it and finally there's a there's another option which is that uh even though you have multiple distributions that are kind of indeterminate uh or or have these kind of intervals where where you know you're modeling things with probability Mass functions probability sorry probability density functions so you're you're just picking out intervals you might think that if you have some process higher order process for example that takes information from multiple processing stages then there's a possibility that all of this is somehow collapsed into each other in a way where you do get actually very narrowly defined uh hypothesis right and again you might think that this would happen in some kind of discrete medium uh that doesn't have to be connected to uh Concepts necessarily uh so so so I think there's plenty of ways in which you can you can get around this problem for me personally I think that uh nowadays I'm actually kind of removing myself from this this debate and I just think that well there those are multiple ways in which you could model conscious experience and then and build models of perception for example and and categorization of perceptual experiences and so on and what's interesting is really whether these models can inform new predictions or new hypothesis that can be tested empirically right yes it's uh we will come I I I we will come to the scientific realism versus instrumentalism thing because I'm sure people want to hear your your take on that um in Greater depth it's I'm really sorry can I can I just add one more thing so so another reason why you just just to kind of again just talk about this you know why would there be one hypothesis that's winning uh one reason for that is nicely given by Jakob Hui and uh Frances Mari in their paper I don't 2020 I want to say I don't remember when it came out yeah I'll have a uh but y but they kind of they they subscribe to kind of Jesse Prince's view of mid levels of the hierarchy relevant for Consciousness but they have actually a very good answer to why there might be you perceptual experience and or experience in general is not kind of indeterminate but is kind of rather fix fixed in a certain uh temporal window right because what's also relevant here is that that when we're thinking about experience we often tend to leave out time but time is a an a really really important aspect of our perceptual experience and our experience in general Consciousness in general we know that there is a certain integration window where you know stimuli that are presented in the same time will be integrated into one percept for example so the reason they give for why there might be a certain uh deterministic quality to our personal experiences and experience in general is that actions

are actually not indeterminate right yeah every action you have to take in your environment is a kind of a determinate situation in which you're putting your body and the way you're changing your environment by acting on it you're moving things around and things kind of can either be here or there I can lift my arm and I can lower my arm right and I'm in between two uh states there might be some gradation between these two states but if I do one thing I can every state or every action I take or every state I put my body into that limits my possibilities constrains them in a in a certain way and my body can be only in one state at a time so when I'm performing actions when I'm planning actions uh I need to kind of experience the world in a way where I don't experience myself doing five things I'm doing one thing and that's actually you know perhaps there were organisms further on if we think that the ban brain hypothesis is true perhaps there were organisms that had a more indeterministic experiences of the world and presumably they would have died out because determinacy yeah yeah exactly yes yeah that's ABS that makes perfect sense I mean again I'm I point people in the direction of Kate nave's work as well I mean she was just on and and she sort of piggybacks off yakob and um Mar's work on this uh sort of in what do they call it sort of intermediate processing right like these are all and again and perfectly chimes with things like affordances um and the fact that perception is enslaved by action to some degree I guess uh well multiple things that come to mind I mean one is it seems to me like a lot of this is uh comes down to the question of attention because we we make the claim that the world is determinate the perceptual world is determinant but I would propose and again this is riding off what Kate thinks that the world is determinant in the very sort of focalized purview of my determinate world now if I have if I'm looking at my periphery it is not determinate at all and I think if to also build onto this action-based picture it's because I can only really I only really Act on what I pay attention to so do you think it's viable to say that you know you're talking about this higher order process which collapses the um probabilistic densities into a kind of you know we can think of it doing kind of a direct Delta function it just like collaps it into a single value is how much of that would be that that is actually just ATT tension right that higher order process is attentionally homing in on a singular you know derives a singular per and then we can say well that's because attention is in many ways the slave of action I think that's very plausible so this is more less The View that we defended with Joe in the fame in the brain paper right so so we kind of put attention in the in the Forefront of Consciousness there uh this is also a view s i I defended a similar view in my dissertation uh and I do think that attention is extremely important for for Consciousness and know yakob actually also thinks that right so he points out that on

predictive or or the active inference stories about uh the brain uh Consciousness cannot be entirely separated from attention because attention is precision optimization right and precision optimization or expected Precision optimization happens on multiple stages in the system happens on multiple stages of the processing stages of the system simultaneously MH so so there's I really there's never a time where the attentional mechanism is completely off so to speak in the brain there's always some kind of biasing attentional biasing in terms of uh expected Precision way weighing of of uh error prediction errors happening somewhere in the system so attention seems to be necessary for Consciousness at least on on the on the active inference predictive processing story uh so yeah I I find Kate's suggestion uh very palatable how much does this um feed into broader questions of agency um teleology if you want to go all the way there and and and relevance I mean I'm interested in theories of relevance or relevance realization John BVI this kind of idea and the idea attention in many ways is the kind of driver or or the product of a organism which needs to uh you know inact relevance realization it needs to be able to disambiguate that which is relevant to it and that which is irrelevant to it and that I think I I've made this argument I wrote this paper it's um currently in review about how I think that attention is ultimately the slave of preferences and I think you could say attention is and that means that attention is ultimately slave of relevance because you know I it's not random what I pay attention to obviously and also the very fact that I can pay attention is is itself a prior an act of prioritization it takes a kind of flattened world and creates peaks of saliens yeah so how much do you think that we can build in values relevance meaning and so on into this kind of more strip back mechanistic perceptual picture that's a really nice question and if you don't mind I would like to read that draft if you don't mind to because this is something that I'm working on now in Brussel with Axel Claremont and actually Axel has done some research years ago on attention and salience in addicts so he has shown he and other people that his Co authors they've shown that there is so so attention is driven by preferences as you said in the sense that addicts do pay more attention to things that are triggering their addiction right so you know if you're a gambling addict and I show you pictures of uh poker chips or cards those are much more Salient to you than kind of neutral stimuli right right right right uh so so so that's the the first point so so I think you're on to something there and I I just want to also say that yeah that's proba like this is this is um evidence coming from humans but if we go to a much more basic level I think that attention is doing basically prioritizing certain information streams over other information streams yeah um and what that means is just as you said organisms that have attention are probably

organisms that live in environments complex enough where they have to do some kind of motivational tradeoffs or they have to tradeoff acting on one piece of information against another piece of information right so there there there seems to be attending to one stimulus against another stimulus so you have to prioritize what you're going to do and so if attention is in the business of prioritizing processing of or trading off processing of certain things over other things then that's exactly very similar to motivational tradeoffs right where you're you're responding to a certain stimulus over another stimulus uh because you know even though one stimulus for example will yield you rewards immediately another stimulus will yield you greater rewards in in a in a window of remove time so so you have and and these two things seem to be quite connected in a sense that if you want to apply yourself to you know pursue a stimulus that will yield greater rewards that are removed in time it seems that it would be useful if you had a some kind of a very riment perhaps system of prioritizing information about stimulus and kind of making you direct yourself away from the from the immediately rewarding stimulus right yeah and I think these things are extremely connected and in fact if we look at the literature on addiction and evidence that comes from rodents so so uh work of people like Kent parage on mice the main driver of what they call incentive salience so so what has been labeled in the literature as the wanting signals so signals which which uh code prefer the preference for stimuli um or the drive for certain stimuli the these uh these signals have been linked to dopamine which has also been linked to to attentional control at least in humans and and other mammals so Precision right exactly exactly so I think that there's a there's there's a lot of uh there's a very strong relationship here uh between these between attention and some kind of uh I'm not sure I I always worry about the word preference because I worry that people will misunderstand yeah what we mean mean by that because when scientists work scientists working on reinforcement learning or you know rodents or or uh some kind of simple animals they talk about preference it's clear that they're talking about preferences uh but they don't talk about some kind of conscious yeah intentional States uh the way we think of them in everyday life and I sometimes worry that especially in philosophy of Mind where people people often come from this armchair position where they think first about how things seem to them uh from the first person perspective for example and then they they look at the empirical evidence I sometimes worry that there might be some confusion and I also I'm a little bit uh squeamish about talking about meaning here because I think that meaning is a very big word um so so I I really like this term incentive salience because I think it's it's connects precisely what you're interested in which is salience so the fact that some

stimulus is Salient to an organism with the the the idea of an incentive so the the idea that the reason why it's Salient is because it is associated by the organism with some expected reward and that's what's driving the organism's Behavior behavior and also I mean inert and overt Behavior so paying attention to something might be a kind of of you know uh covert uh Behavior right yeah yeah yeah considered conceptualized as a kind of covert Behavior right yeah I I certainly view it as such a sort of mental action uh yeah exactly down exactly the same sort of um prob you know the same base graph if you wanted to you know the same c and e Matrix if you wanted to uh yeah so there's plenty of things out I mean one is you're absolutely right I think it needs to you know we always need to say we're talking about preferences in the basian belief sense of that neurons have preferences the atoms have preferences but they they have basian preferences they have attractor sets they have so on or at least they look as if they can be described as if and so on and we've had this and we'll we can come to this of course um when I talk about meaning the reason why I do so is because I think the the way that preferences in slave action is is encoded in a hierarchically deep fashion such that you know the certain things that we can claim have uh make our lives meaningful are those which seem to have this trickle down effect such that they one way of thinking about it is in terms of meta stability you know the the sort of the deepest wells in which the other attractor Wells are kind of subsumed so I've been thinking about this in terms of okay you know maybe my ultimate you know what makes my life meaningful is you know for evolutionary reasons is being a social adaptive member of society now what I pay attention to although it's indirect and and implicit and covert is often a manifestation of that so why am you know why am I writing a paper because I want to get a PhD why I don't I want to get a PhD because I want people to think I'm impressive what and eventually in some sort of Aristotelian fashion you ground out for me in these kind of hyper priors which seem to have this kind of deep filtration into our very low-level attentional schema one question that I've been thinking about as a kind of followup is I think it's very reasonable to say that you know when I'm hungry because it's a very deep rooted prior it kind of has a a standardly high Precision waiting that prior such that if when there's deviation well maybe not maybe it's a um when there's deviation from that prior there's an action policy that grants it high Precision waiting I should say because we're not always hungry or aware of our hunger levels so there's a slight Nuance there um but you know we can just say that it's a you know for lack of a better word it's kind of an import deep prior MH I think it's fair to say that when that's the case I'm obviously I'm far more drawn to food right of course I'm drawn to the fridge I'm drawn to

adverts of food now but I've been thinking is that is it the case therefore that preferences and and that which is afforded in terms of basing preferences or high Precision weight at that moment literally change our perception in such a way as it's sort of like a sapier Warf hypothesis for V for perception or is it the case that our perception is uniform and what ends up happening of course our attention isn't uniform but our perception is uniform and what ends up happening is we kind of overlay uh our perception with the kind of conceptual schema and well also kind of the attentional schema but actually in the sort of fundamental aspects the intrinsic aspects of our experience uh the colors the sounds and so on things remain the same have you got any kind of intuition on question uh this is a good question because I think uh so so I'm kind of torn on this on this question so so if you're looking at these hierarchical if you're thinking in terms of these hierarchical basan models and you know active inference where there's cascading predictions constraining the activity on the levels below it seems like there should be some form of influence of you know your Global States for example over your local States right so so being hungry is a kind of a state of deprivation which seems to be a kind of a state which is I would I would maybe label it as a global state in the sense that it pertains to your organismic functions and if it's not satisfied you will eventually die right so so it's kind of a a Glo a state of global importance to the to the cognitive system so to speak because it's it's the existence of the cognitive system is contingent on on uh on keeping that state within certain bounds so and then there's a question of like whether there's a way in which through as you said saliency maps and attentional modulation and all these things and as you said conceptual overlay for example there's a possibility in which this higher order well let's call it Global State again would constrain the space of possible hypotheses or possible models in such a way that it would have determinate effect on your conscious experience um and I think it definitely can have an effect on your conscious experience overall because if you think of your effective experiences yeah you know we have the term hangry which clearly indicates that you can your effective state is changing and and you can go into a certain mood because you're in this Global state of deprivation and then your your fuse gets shorter and you get irritable and so on whether it changes your perceptual States I think that under some models people should be committed to claiming that it does I am not so sure about that because this is and and then we're here getting to the to the second kind of uh part of this dilemma for me because there's this whole literature about cognitive penetrability and whether our beliefs influence how we perceive the world qualitatively whether they beliefs influence perception and from my understanding um for example the work of Chess firest Stone and Brian Shaw

the evidence for that is very very limited yeah so it seems that our beliefs can for example influence our actions very easily yeah but they do not change our perception as readily as uh it might seem intuitive and so here I would probably try to be a little bit cautious and now with one curve yet and that's that's something that I'm working on now which is this um the effective ladness perception so as you know and as many people at this point might suspect being a student of D denet I'm no fan of qualia and phenomenal properties seem to me you know metaphysically suspect and I would like to exercise them or at least deflate them as much as possible in our ontology so I think that a lot of problems related to ineffability and a kind of intuitions of phenomenal States being some how mysterious comes from the fact that in our perceptual experience for example let's take Vision as a as a simple experience a lot of information gets compounded and compressed in a way where we are unable to track all the information that's being you know uh packed into into a perceptual representation by our perceptual systems and so um so you know you said that so one example is that colors appear to be intrinsic properties of objects right to us even though we know that they are actually bundles of complex relational properties that depend not only on the surface properties of the object but also the the source of light uh viewing conditions context our past experiences uh the state of our ocular system and also how our visual system is processing it right so there's a complex Nexus of conditions that obtain outside and inside of us that make colors appear in a certain way H and yet all this stuff is kind of flattened into this perception of colors as properties of objects that are out there in the world and I think that so that's part of the story but even if we if even if I present to you you know the best state-of-the-art knowledge about color perception you will say like yeah but there's something ineffable missing there's this qualitative Gena of of of phenomenal experience and I think that this is perhaps because in our perception there are Pathways for example to subcortical areas which might be imbuing our perceptual experience with as you said some kind of slight emotional sence right or something like that so what people have called in the literature as micro valances right yeah yeah um so I find this picture attractive especially I find attractive the idea that somehow affect might play a role here but I think that we have very at this point in time at least we have very limited uh very limited evidence for that so this is something that we're doing now in Brussels trying to find out more about uh the effective ladness of of uh perception and also it's worth taking into consideration just finally kind of as a reconciliatory note the fact that you know even if perceptual States as such sensory states are not penetrated or somehow imbued with effective or other types of information from other area brain areas or brain processes that doesn't mean

that our perceptual experiences as a complex cannot bring all these things together right so the perceptual experiences are extremely complex so it's not just about sensory properties Drey has for example written a lot about you know how Concepts structure our perceptual experiences so there are sensory properties in our perceptual experience but it seems that there's much more happening yeah than just the sensory properties alone and that's something I'm happy to concede and I think is highly plausible but the claim that you know I see the the candy bar rapper as brighter because I'm hungry right that's I would I would I would be very cautious about this type of CLA you see that that that's the kind of paradigmatic example right because um I think you're absolutely right we need to bracket off the domains in which perception might be from which perception might be influenced so affect is clearly permeating perception uh and and clearly uh the preferences or the PRI or the hypotheses are affecting that right as you say h hungriness is a perfect example such that there is this prediction error or free energy spiking with respect to this one homeostatic Prior and that permeates my experience that becomes a uh you know a kind of just all pervasive um yeah uh experience the other thing that clearly in some the wrong is the wrong word pervasive but is over you know intrinsic to my experience is the attentional modulations the the rivets and the peaks of my prioritization and clearly as I say and I outline this paper preferences are shaping that um such that I'm hungry the candy bar is going to draw my attention I think this is fair enough to say more than it would when I'm full right like I had this experience yesterday I went out for lunch I was really I was with friends I was really hungry and there you know news you know news agent shops or whatever selling food and you're in that state of hunger drawn to that and then I leave and I'm like I don't care I don't want to see it but yeah exactly the point with me does that actually alter the fundamental character you know uh the the the qualities of that experience now I think an interesting contribution to this question is psychedelics because yeah there's something about psychedelics which if we take a kind of Rebus or alus approach loosens those priors at at some level right I'm not saying the the sort of preference priors but the priors over let's say um I don't know like it's kind of like in dreams right like you might have a prior that um I can't sort of dissolve through walls I don't know like this clearly there's probably no but something like that and clearly something in like dreams and psychedelics that gets loosened such that in perception I seem in the dream to be able to sort of blend through walls and and so on now there there there does seem to be a relaxing of a mechanistic prior that permits um changes in my visual experience in this case I guess the question here is slightly different is there could there be changes also in my sort of preference priors such that if I was uh if I was

super hungry and then took siloc cybin would it literally change my visual of uh perception or is it just that the mechanistic part of that is changed so there's kind of two different priors or preferences that we need to be talking about as well can you can you sorry can you repeat the last part so if you took silos cybin it would change what because I think that you lost me there at the at the very end so I guess what I'm saying is you know the way people talk about well the Rebus hypothesis relax on psychedelics is somewhat vague about which it's a kind of narrative account of Prior it just relaxes and I think KL has said this himself I'm sure Robin would as well but there are different types of priors so one would be a kind of mechanistic Prior okay what do I expect to appear in my perceptual experience so I do not expect uh you know two things to be in one place at the same time I don't expect for things to move through solid objects yeah but then there's also the kind of what you know you could call phenotypic or preference priors which is defining of the organism I expect uh to be satiated I expect my blood glucose level to be a certain level I expect my temperature to be a certain level now seems to me that what siloc cybin is doing is relaxing those mechanistic priors such that you can start seeing things go through walls and the same in dreams and whatnot but I don't know how strong the evidence is that it changes the preference priors right does it it would so my thought experiment was if I was really hungry and then took siloc cybin I can imagine it would hijack my attentional apparatus such that I would start homing in on all the food but would it literally you know would my steak start like glistening would my candy bar start I don't know I guess at that point it's kind of hard to say what what part of the system is being altered but that was kind of the the fault experiment okay okay I see I see now um yeah so well it's clear that psychedelics can induce sythetic experiences right so so they can alter how we perceptually experience the world dramatically and uh they can make us experience you know certain uh this kind of modality mixing States right so so you you see the music for example stuff like that so so that seems to be a very deep uh change in in how you experience the world in a sensory way mhm so that's one thing um I totally agree with you about the mechanistic priors and the fact that this seems to happen uh on drugs a lot um and that there is yeah that that that that as just as you mentioned as in dreaming like they seem to be relaxed so you're entertaining hypothesis that would otherwise be outside of of the hypothesis space that you could entertain and what's also interesting about dreams and hallucinations is that at least from a philosophical perspective that philosophers often kind of create this stman idea of hallucinations as something that's you know you're you're suddenly in a state where you don't know that you're hallucinating and you're confusing hallucination for

reality this is actually extremely rare given given current evidence and it seems that people under the influence of psychedelics very often are aware that hallucinating just as sometimes people in undergoing lucid dreaming they can enter a lucid dream by re realizing kind of having this moment of aha of oh my God this is completely unrealistic yeah um which is in itself also interesting because that means that you you you're relaxing a prior on certain levels of the hierarchy or in certain parts of the system but at the same time there might be aspects of the system that are still kind of monitoring what's going on and are able to say this is this is not reality this is we're we're on drugs now right um that's one thing and then finally going to these preference or phenotypical priors I think that's an open question one thing that's for sure is that and I don't know anything about literature uh that speaks about this change in these type of Prior in preference prior during these psychedelic experiences but there is a lot of evidence that psychedelic experiences do change people's preferences and behaviors long term right so so there's this famous effect that people who have undergone uh a psychedelic experience are for months afterwards scoring higher on openness and lower on neuroticism and and and these kinds of personality traits which would I would at least correct if I'm wrong I would ascribe to a change precisely in these phenotypic priors where sure you're kind of changing your behavior or your attitude towards the world and people yeah uh because of that experience so perhaps within the experience it's also possible now you know whether whether there would be this permeation of perceptual or sensory experiences by your preferences that's a very interesting question and uh I mean we're in the kind of gold an age of psychedelic research these days so I'm sure that sooner or later someone is going to control the conditions of their experiment in a way where they will be able to show that this happens or it doesn't happen perhaps right well I guess yeah it's it's it's super important to unpick exactly what the hypothesis would be in that in in in that experiment because as you say clearly as you yeah I mean there's overwhelming evidence that of course you get these longer term effects and then modulating perception and conscious experience in interesting ways but again the question is still there which is okay what part of your conscious experience is it uh modulating so okay you're more open you're less neurotic it's going to be changing your attentional schema it's going to be changing your effective schema right the things that used to piss you off are probably going to piss you off a little bit less you might be less reactive so you might have more space between the the percept and the response but is again I mean I mean the really really the kind of fundamental question here is how much of conscious experience is there beyond all of that you know like how much is color these kind of supposedly intrinsic things

that are not um overlays even though we know mechanistically they might be overlays for sure like to what degree do are they are they just part of all of those things I mean the evidence kind of the my intuition is that they're not because I could pay attention to something but that doesn't change it it changes the way that I am living with it right I same with affect it doesn't change I mean I don't know it's hard to say you know no no no no I this is a this is not so this is not uh an argument that has not like this argument has been made before right so this idea that when stay off intuition when you're paying attention to something it changes your quality the quality of your relationship to that thing yeah it's like with a camera you know you're looking through a lens of a camera and you're adjusting the focus but it doesn't change the object or the contents of of what you're seeing right so so people have argued against kind of predictive processing stories that attention might be kind of uh narrowing the the distribution right making it more precise but it's not changing fundamentally what you're perceiving right it's if it was red it's still red yeah yeah yeah and [Music] um uh yeah and and so because I guess the thing here is it's not changing what you're saying I mean again like we kind of have to distinguish between the numo and the phenomenal right it's not I mean I don't think anyone would argue that it's like unless you think that you know you have some quantum mechanics and you think there's Observer like Park calling of that and you Tony the mechanics of course if I look at my computer I'm not it's not sudden apart from the very sort of potential Photon changes I'm not morphing it into something that it wasn't sure but phenomenally the as you say the contents of my experience I don't know I'm a little I'm a little L I'm a little less secure on that front um I mean I'm just noticing things that I didn't notice before which in a sense is changing the content but whether it's changing the content as it pertains to the object itself is a slightly different question right so for example uh the yeah so so one way to think about it is so so okay so we have to kind of I think separate several things yeah so one thing is that uh my claim and the claim you know that other that that Joe has made together with me in the in our paper is um attention is kind of modulated is is is corresponding to what what block labels Act Consciousness right so so so attention is not supposed to give you a full account of phenomenal qualities that's that's one thing um and I think most people most higher order theorist theorists agree with this that you know you can probably do more than you expect with just this kind of story about introspection or some kind of higher order lower order relationship so people tend to think that you cannot get any quality postive information there that's not entirely true uh and um uh Andy Clark has actually a very good paper I think from 2000 or 2003 called access implies uh access implies quality I don't

remember the title now I'll find it uh it's a very nice paper is it a case where access implies qualia yes that's 2000 yeah uh yeah so that's a nice paper also that shows that the distinction between access and phenomenal Consciousness might not be as clear-cut as block and his supporters would like but just to be clear on our picture attention is just you know this kind of probing mechanism in the netian terms it's it's this access to information mechanism yeah and now so that's one thing uh now the second thing is as you said that um that the the claim about whether it changes the content drastically whether paying attention to something changes the content of your experience and here again we have to be very careful because something can change uh your experience can change in some qualitative ways without the content changing per se so so Drey has this very nice example where he Compares two situations one in which I show you a piece of paper on which I drew 30 dots and I you know I just kind of show it to you like this it off and then in another case I show you the same piece of paper but there are 31 dots so technically speaking technically speaking the number of dots has changed between the two pictures so the content of your experience has changed Technic at least that's the point that he would like to make in my in my view unless you notice that there's an extra dot the content of your experience has not changed right and it's only through sharpening of attention Could you actually notice that there was a difference right in your in the between the between the two percepts so so so I think that there are these cases where indeed uh paying attention can change the quality of your experience even though it might not change explicitly the contents of your experience interesting that's an division right because I think most would view those as isomorphic they would view contents of experience and for and the phenomenal qualities of experience as isomorphic yes so so I think it's useful to maybe think of oops I think I'm getting now sun is shine into the office uh so so you might think about this in terms of modes of presentation of content right so attention seems to attention is not only picking out what aware of but it can change the mode of presentation of what you're aware of to some extent just as I said earlier with the metaphor of sharpening the the the focus on the camera lens yeah the the the subject in the picture doesn't change but what's changes is kind of the way you're capturing the picture with your camera and I think this is this this might this kind of talk of modes of presentation might be useful as a as a useful kind of intermediate step between the the content itself so what is represented yeah the way it's represented right um this mode of presentation those that that distinction might be might be useful here as well do you think there's a strict do you think one can draw a strict line there between contents and mode is that do you think that's feasible I mean is there a kind of taxonomy of modes in which things can be uh perceived or

can my mode in some ways become opaque and become a content I think the latter can probably happen so so I there's there's not many people in philosophy of Mind these days talking about M of presentation so charers is what is I guess the the main figure when it comes to that uh Hilla Jacobson talks actually about something similar when it comes to effective uh effective coloring um and uh some other people do as well but it's a very small group of people these days that talk about modes of presentation as distinct from contents but I definitely think that there is a there is a there there are cases where we conflate the the way we experience things or the way they are presented to us in experience with what they are so I think I think so hila Jacobson has for example a very nice uh example that she uses in her paper as part of her argument which is that the same object the same food item can look delicious and disgusting to two people yeah and so what seems to them they are looking at the same thing right so so the sensory properties do not change presumably or we can even assume that the sensory properties don't change between their two experiences but the effective mode of presentation of these two sensory experiences can be dramatically different right so so and and a simple example that I used in my talks of hijacking Hill's argument is is a piece of licorice because people often think that you know people can have strong reactions to different food items and licorice is a is a piece of sweet that almost or Marmite would be another example those are two things that people have very very strong opinions about and they have very strong reactions to but the a jar of Marmite or a stick of licorice are not by themselves repulsive to us people just perceive them as disgusting or delicious and you know they can have this kind of different ways of engaging with that content so maybe that's another way of thinking about mode of presentation as as this kind of way in which you engage with the with the content of your experience right what's uh Shakespeare there's nothing good or bad but only thinking makes it so yeah yeah that's that's Hamlet um yes I mean Shakespeare I guess wasn't thinking about whether yeah I guess uh I would have of course I have no issue with the idea that licorice might look words Fair here might look disgusting or look attractive to people but I guess the question is do they look disgusting and look attractive to people are we overlaying that or is it suffusing our I mean even if it's not overlaying it might be suffusing our experience the question there as we were talking about is are the sensory properties being altered but I guess I mean I guess that's a I guess it's an experimental question ultim I mean how you could how you could do that without qualitative report I mean that that which might be quite limited I'm not sure because again I mean language fails us here so this is what we're interested in here in Brussels right now and and one way you could think about it is well do

you see changes uh in the sensory processing areas right when you're when you're putting people in a brain scan right do you see changes in activity depending on the valence of the stimulus that they're showing them so so a common stimulus used in experiments on uh affect are smiling and unhappy or frowning faces or or angry faces right so you show them two types of faces they distinguish whether they are happy or not and then the question is where do you see the difference between the happy and the and the frowning faces right is the is the difference represented the difference represented in some area of the brain that's associated for example with processing value yeah uh some kind of one of the frontal areas or is it maybe or whether we can find the differences already on the level of perceptual processing so this is something that we're aiming to uh to test here and there is there's kind of the the available uh evidence is kind of equivocal I would say so there are some papers that show differences in perceptual areas there is also some work that's been done in Nicholas Krias's lab which seems to show that there are actually it's not that the differences are represented in sensory modalities but there is a sensory modality specific coding for the valence of the stimulus so you could think that there's a universal code for Value in the brain but in order to compare where you have to bring two different stimuli right so if I if I ask you if I show you a happy face and a monetary reward something like that if I give you two two different types of stimuli if you're comparing if you have to do some kind of a trading off between them um you have to bring them into some kind of a common neural currency yeah so there's K. O'Leary's lab uh has shown that there might be this Universal value coding but before items are brought into this Universal value coding for comparison there's also a a modality specific coding of value uh in the brain that is not sensory per se but there is there are differences in how values are coded uh depending on the modality from which the information comes which would seem to be this kind of intermediate state where your perceptual experience might be uh effectively Laden even if the sensory properties stay the same because what's there's something that's being added you know on top of sensory properties it's not changing how the Hue and the brightness and the color that you're perceiving but that there's something else that's overlaid on that as you suggested earlier that is changing the character of the overall perceptual experience and this is precisely Hilla Jacobson's View and also Frederic de Vries has a recent paper where she also adopts a view like that so much to read so much to read I'm writing down notes I mean yeah I mean I'm very impressed I mean it's funny that we still haven't resolved these issues uh since 1983 when Jerry Fodor wrote modularity of the mind I mean he he he would if he was still here he would have loved to um he would have loved to have uh listened to this conversation

because I'm sure he'd have lots of opinions but perhaps or he would be maybe he would love to or maybe he would be extremely frustrated you know yeah well I'm sure I'm sure again I did I never met Jerry of course but I'm sure he would uh whatever it was I heard he was very he was an opinionated man I mean for good for good or for bad okay cool um that was that was that was super interesting I mean I love that I read modularity than M when I was at Uni and loved it I mean I loved it as an intellectual Endeavor I mean I had no skin in the game but um I thought it was I thought I also think there's a lot you can learn about writing clearly and precisely from from foder and from Drey so I've never read any Drey I mean he's come up over and over again I had a little foray into I read one of the like radical and activist books um okay can't remember who wrote it and there's a lot of sort of dryan and milikan sort of critiques as in they're critiquing them because they don't think that any you know only represent the only representations are those which are socially scaffolded yeah oh so it's the second book by uh Dan May yeah and Eric mine yeah the the basic Minds evolved I think yeah I read I read yeah I read one of them um and again like it's not really my argument um but it was interesting enough I should read milikan and and dry but life is life is short high but they're highly recommended so so I really um knowledge and the flow of information especially if you're working on predictive processing and these topics I think it's a really beautiful book that probably is the first philosophy of Mind book that took information Theory seriously uh even though there are some distortions and he at some point develops his own kind of you know version of of of Shannon's communication Theory kind of Mong view in the end but but it's a beautiful book and it's an extremely funny fact that that book was published in 81 I think so 30 something years after you know Shannon's commun mathematical theory of communication became a thing and this kind of shows you how long it sometimes takes for philosophers to you know catch wind of what's happening in science did um this is a kind of historical question did Chomsky ever reference Shannon was try do you know if trky was aware of Shannon I'm I think it would be strange if he wasn't given that he was you know where he was working and that would be surprising to me however I don't actually know so so I've read Chomsky uh but a long time ago like his linguistic writings a long time ago and I mostly the last things I've read were mostly about um about mental representations oh really yeah so so he has a very very deflationist view of mental representations you can probably find him you can ALS Francis Egan in some of her papers okay takes his minimal view of what she calls erat's representationalism as a kind of uh Target yeah I did uh I did Linguistics as my undergrad oh okay so I I read Chomsky from you know um syntactic structures all the

way to minimalism uh but this was all very sort of biological Linguistics about in you know the question of whether Linguistics there you know a innate linguistic device or things like this so slightly different but uh yeah I was just wondering because obviously he was writing in you know the 60s 70s so yeah yeah okay uh on the question of representations I mean actually it's funny because both people I've had on who are expert in representations you and Alex kefir have both recommended me dry and I'm Alex's work on representation is great so I am a of his work you I really liked your uh moderate predictive processing paper it was challenging because um I feel I it kind of made me realize like oh I've been taking the vanilla predictive processing because I'm not personally in my own writing I'm not wedded to any particular architecture whether it's discrete State space continuous State space I take the vanilla position I run with it and so on but it made me think like oh there's so much more for me to learn exactly about predictive coding predictive processing schema well I found really interesting was this idea of well this ambiguity more broadly in the literature about what is being represented at different layers of the predicted processing hierarchy so you say and can quote um there's an ambiguity between levels which perform the function of standing in for features of the environment uh and those which act only as meta representations by standing in for the features of levels Downstream so there's this idea that obviously the high you know the higher you the higher layers are predicting I mean all layers are predicting but you know they're predicting the layer beneath them and the prediction area is rising up and there's kind of this meeting in the middle and this Hier this horizontal explaining way of prediction error but I thought you this is a really nice observation which is where does the external world come in right because there seems to be this equivocation where we say okay we're predicting the external States across the mark of blanket but then in this predicted processing scheme we're actually only predicting the ne the layer below and so there might only be one layer which is actually receiving the sensory input so where where have your where you know this was 2017 you wrote this paper where have your thoughts now evolve to with respect to the architecture of processing uh yeah so just on that quote right so so just to be clear about what I was kind of attacking there so so what I was attacking is this ambiguity between in in an in how people answer the question about what is the function of of a layer in a in a predictive coding hierarchy so is it representing the world as some people will say or is it predicting what's happening just below on the level below and a lot of people when I first encountered predictive coding and uh it was very informed by Mark SPAC who was my master thesis supervisor in edinb and he the way he presented it to us initially was uh in saying that the

brain is just minimizing the the story is that the brain is minimizing prediction errors and that's all it cares about and then it turns out that the best way to minimize prediction errors is actually to represent the world right so so so to kind of have the the the brain become a kind of distorted of course but a mirror of nature in some ways and that this is this the reason why predictive processing offers a a kind of a story about the organism building up its own model model of the world is because by building the the the process of building up your own model of the world is just the best strategy for long-term prediction error minimization um and so my problem in that paper was that different people want to clink to different interpretations of what a single layer in the hierarchy is doing depending on what on what serves their wider uh wider philosophical Pro object so people who are anti-representationalist will say like oh you know the function of a layer is just to predict what's happening on the layer below uh and other people will say like the function of a layer is to represent the regularities in the environment at a certain level of abstraction and at a c in a certain time uh window and both of these interpretations can be true according to predictive processing it's just that which one you're try you're emphasizing seems to depend on your uh external let's call them or at least external to the framework philosophical assumptions and convictions and this is why when I wrote that paper it was already becoming apparent that for example Carl uh is uable to different interpretations of predictive processing or predictive coding or active inference whatever you want to call it depending on who he's coauthoring a paper with so so if he's coauthoring with an an activist he will take a more an activist stance where he will uh try to avoid talking about internal representations or or the or building up an internal model of the environment yeah when he's writing with someone who's uh coming from a more cognitivist fodorian uh persu with with a kind of fodorian persuasion then he will lean more into the representational talk so it seems that uh that there was this ambiguity uh about what's what's happening in the framework and today I think I'm kind um so I used to think that maybe there's a definite answer to this question which interpretation is correct today I'm actually thinking that uh the semantics maybe are kind of disconnected from the from the formal layer MH so I have this I I'm I'm at least when it comes to predictive processing I think there is no definitive answer whether we should think of the generative models for example as representations or not I think that this will depend on your other commitments about what a representation is for example uh on the wider goal of what explanatory project you're engaging in I think those are the things that will actually determine which inter interpretation you will apply to predictive processing but I'm not uh really committed to to either of the

interpretations and um this goes a little bit back to attention in the sense that I think what you attending to in the predictive processing story might influence how you will want to interpret uh the framework I'm sorry if this is kind of a oh no I love it not satisfying answer you know it's maybe I'm the same I'm the same uh despite being a host and and I feel like part of being a host of active inference podcast is in some ways you should be wedded to certain ideas I'm pretty ambivalent about or not even ambivalent I'm open right I'm epistemically open I don't know um that might frustrate some people but I don't know uh and also this isn't really I don't really have an area of expertise yet so so I appreciate the I appreciate the epistemic humility I mean it's an open qu I mean I mean fundamentally one of the issues is no one really knows what representations are I mean like what what what the I mean as you say all philosophical the all theories have their miracles they have their axom right so are you a dreki and are you a milikan follower are you you know do you think that there needs to be a you know a strict isomorphism between that which is representing and that which is represented and so on and so forth I wondering whether even if we park the sort of what is being predicted whether it's the lir below or whether it's the world what are the sort of unifying features therefore that we've come to in terms of a predicted processing landscape so for example there's this idea that the priors of one level are the posterior furnished from the lay above yeah now what does that really mean is that is that what the is that what people talk about when they're talking about predictions coming down the hierarchy and and I guess the question here is and I said this to Alex as well I think there's a real I mean I struggle visually conceptually to really imagine a predictive processing hierarchy because I always view it as in time right dynamically uning such that you have these I mean at a very vanilla level fine prediction eras coming up predictions coming down and they get canceled at some point you get a hypothesis woo the world appears like great but when you actually start thinking about okay mathematically formally what are the architectures that we're setting up in place it becomes very tricky for me to imagine because I think of a layer that is simultaneously predicting and being predicted and explaining away prediction error so just at a very broad level is there for myself and for other people listening and watching a way of like formally disambiguating what might be happening or should we just say this is some dynamic system and you can kind of get by with the vanilla approach I think you're putting me in a position where which is you know way above my pay grade here no but um uh yeah that paper you that paper you do cover quite a lot of technical round a little bit but I must say that also I'm not entirely sure how well you know my technical grasp has so so whether I was that correct there when about in the

things that I was writing did anyone send you an angry email no no never never but this this I I'll answer that question your question first and then I'll tell you one anecdote about this paper so um so yeah so you're right that there's a lot of variability when it comes to how you can implement the general predictive coding scheme and uh Michael Spratling has a very nice paper from 2007 called an overview of predictive coding algorithms where he compares different first of all he gives a kind of a brief historical overview of linear predictive coding and where it came from and then he talks about the three uh yeah the three major varieties of predictive coding as applied to hierarchical networks which is ran Bard's original model uh Carl's uh free energy minimization model and his own predictive coding through biased division modulation I believe that's his the name of his algorithm so he compares these three algorithms and Michael Spratling is actually a guy from London who who has done uh a lot of work on modeling uh neural populations using predictive coding and building very large neural networks performing different cognitive tasks by through by implementing implementing predictive coding and so uh one of the problems that that paper highlights is that because there are many different ways of doing predictive coding you know it's not stable from one algorithm to another what the prediction error is right and you know what is propagated upwards what what is propagated downwards even how to delineate a layer of the hierarchy well quite yeah uh so so there's a lot of and that's what I also touch uh that's what I also touch on in that 2017 paper moderate predictive processing so there seems to be a number of possible ways of doing predictive coding and depending which model you will or algorithm rather I should say you'll pick there will be certain differences and what counts as what so in the end when I was writing my PhD I actually decided to go to a higher level of abstraction and I looked at the work of Iris vanroy and Johan Kout from Radboud University in Nijmegen in the Netherlands so they have a bunch of papers in which they formalize coding on a more coarse grained in a more coarse grained way as uh by using basically Bayesian graphs and imagining each layer of the hierarchy as a Bayesian graph with certain dependencies and uh you know connections between variables so in their model they assume that each layer of the hierarchy very much like in distance models each layer of the hierarchy which will have a set of hidden variables and a set of prediction variables right and these uh these these variables will correspond to uh yeah so so these two types of variables will correspond to kind of the the latent variables and the so-called observed variables in a way where the the latent variables are the are the hidden variables that get inputs from the observed variables on the level above when predictions are generated predictions are generated uh and this will be integrated and then at

the same time will'll issue predictions so so will influence the state of their their own observ observed variables and these will kind of cascade downstairs so so they imagine this kind of simple probabilistic graphs that are connected uh one to another and they kind of operate in this kind of two stages right so they either do inference or they or they um observe what's happening and I think that's a very uh General this is probably the most General way in which you can think of predictive coding or additive processing because it leaves out error units so they don't talk about how errors are propagated into the system where the where the errors should be connected to to things and stuff like that and that's I think really an important abstraction because the way errors are processed are dramatically different between different predictive coding algorithms and their very abstract and very general way of for normalizing predictive coding has the one crucial benefit that it abstracts away from different algorithms so that you could still capture the general similarities between these algorithms without talking you know or getting bogged down from my perspective into the minutia of how each of the algorithms is implemented exactly now there's a trade of in going via that route because by going for a higher level of abstraction you're of course losing detail and if you're losing detail sometimes you might actually find out that you cannot disambiguate certain claims about predictive coding on that higher level of abstraction so people who have for example very demanding views of mental representations as structural representations that are capable of action guidance and error correction and stand in some kind of homomorphic relationship to their targets in the environment well it becomes very difficult to disambiguate whether these abstractly captured predictive coding layers are working as structural representations because you've left out so much detail on the table that you're just not able to exactly adjudicate whether they fulfill all the criteria or not right so so the fact that an abstract abstracted and idealized description doesn't fulfill these criteria that these people will have for mental representation doesn't mean that a particular model like spread links Network or distance network wouldn't fulfill these criteria uh on its own when it's properly implemented with with a greater uh level of detail that's again not a very satisfying answer I feel no look I no I mean I'm glad personally that it other people find it confusing and that I can go to spratling or Carl what remind me of the two Netherlands the two Dutch people who you were influenced by Iris Iris vanroy Iris Iris the way you would write it in English van and Roy is rou i j yeah and Johan kout okay perfect thank you so highly recommended uh very good very good Visions no that this is I like because I I think part of um part of my role here is to point people in certain directions right um so I'm seeing here cognitive cognition and intractability so they they they

they worked a lot on cognition and intractability I think um the paper I'm thinking of or the where the formalization appears is something like to be precise the details don't matter oh there's free energy minimization and information G the devil is in the details might be that yeah that's I think a commentary on Carl's paper where he mixes discrete and continuous variables so I'm thinking of uh 2017 Johan kout Harold becker and Iris vanroy to be precise the details don't matter unpredictable processing precision and level of detail of predictions that's in brain and cognition okay fantastic well thank you for that uh it's a paper that definitely influenced a lot of my thinking about uh about attention in the in the excellent I mean it's a sh framework it's a shame I wrote this paper before um it's difficult to keep up with all the literature because also the literature has ballooned so I started my PhD in 2014 and at some point I was just getting like there were 20 papers coming every month on the topics that I was working on just from people working on predictive processing so you know there were in five months I had a 100 papers to read which is kind of yeah yeah unmanageable right so so so um some point that literature just exploded I'd like to know your anecdote about this so so the anecdote since this is you know a public facing podcast or at least I hope hopefully the the academically facing podcast the the story about this paper is that I wrote it uh I guess it came out in 2017 It Was Written somewhere between Yeah March 2017 it came out so it was written mostly in 2016 so the early days of my uh PhD in fact and um and it was a little bit you know it was hot on the heels of Andy Clark's radical predictive processing which is which the title is supposed to be a reference to and the debate between radical and conservative predictive processing so precisely uh representation light or representation heavy interpretations of predictive processing and I made a terrible mistake when naming this paper because it's called moderate predictive processing and to me at the time it was very clear what the title refers to and what it is about yeah um and I thought you know it appeared in this collection of papers philosophical papers on predictive processing edited by Thomas metzinger and Vana visa and so I thought like everyone will know like this will be obvious that if you're reading like if you're Googling stuff about mental representation and predictive processing people will just find this paper no problem and it actually turned out that because the title wasn't immediately informative people just didn't know what this paper is about so I was meeting people in you know four years later or three years later who had no idea that like this paper came out and they were I was sitting in talks and people were talking to me about the these things and they had no idea that this paper came out and at the time it was one of the very few papers about mental representation and predictive processing but because it was just badly named just

people it didn't show up in searches so I would advise and I would advise I would advise especially Junior people to to be mindful of how you title your work yes because that actually can change the discoverability of your work quite significantly yeah they prob they thought you were making a political point they thought you were H very good yeah no you're absolutely right just like fill the title with buzzword right just free energy free make sure the word free energy in there or active inference um yeah just yeah well discussing public facing work people are going to be surprised if I don't mention the emperor's new mark of blankets but I don't want to spend too much time we've been going for a while as well and I'm aware you've probably got dinner I've got dinner and so on but it's sort of your I mean again this is not this was not your solo project um this was actually I don't know whether did jelly brunenberg lead this or was his name just first up uh so the way that project came about is because y came to bom where I was at the time just fresh after my PhD and he gave a talk about Mark of blankets uh which is also by the way a topic which placed prominently in the previous paper we discussed in the 2017 paper about mental representations where I talk about the insulation of the mark of blank with a mark of blanket between different layers of the hierarchy so after my after my after I finished my PhD y came to B and he gave a talk with his where he was just trying to understand the difference between Mark of blankets as they're described in the literature on probabilistic modeling and and in literature on basian inference and how they are used and described in free energy modeling and so he was very confused uh I was very confused and so that was a paper that he was that was a talk which where he credited it to Manuel baltieri and himself that was something that that was a project that they were discussing and working on he gave the talk and I later told him like y I love this project because I've been trying to figure out how they understand Mark of blankets for a while now and it was something that I kind of bumped into during my PhD and I it wasn't entirely clear what is the correspondence between the way that friston describes Mark of blanket and and how Pearl describes Mar of Blancas during my PhD I bought Pearl's uh 1990 book I believe u on probabilistic inference in thinking machines I don't remember the whole title but it's the book in which he which is often credited as the source in which he introduced the notion of a mark of blanket so I bought the book I read the whole book and I still didn't understand how it's supposed to gel with the free energy stuff and then for half a year or something like that you know I was like okay Y and Manuel wrote the paper it's all good I I I don't know half a year more probably passed and and I talked to I was talking to Joe and we were talking about something completely unrelated a paper that Joe was writing at the time which also had some

probabilistic modeling elements in it and we talked again about Mark of blankets because we were talking about this topic and uh I said hey I got a great reference for you because I'm I'm I wanted to write something about this but Y and Manuel I think already wrote a paper about this so I'm going to ask y if he has a draft and I'll send it to you and so I contacted Y and it turned out that they haven't written the paper that the paper kind of you know they they they didn't have the time to to figure everything out it kind of laid in the in the in the in a that was locked in a closet so to speak where it was kind of Sitting Waiting for a better time and uh and somehow we just proposed that hey maybe we could collaborate on this maybe we could help you bring the paper back so we discussed this stuff and uh uh we started writing a paper then we started thinking okay so where should we where should we try to pitch this paper like what Journal are we writing for and I think it might have been me although I'm not entirely sure who said you know hey how about BBS like because it's it's it's it's getting a little bit annoying these people in the free energy literature they're stting they're starting to claim that a bacterium has a mark of blanket a brain has a mark of blanket a person has a mark of blanket the Earth has a mark of blanket right there are all these papers about the biosphere or the climate as a mark of planetary Mark of blanket and so maybe given how important and how prominent Carl's work in computational neurosciences these days maybe we can pit it to BBS and so we wrote a whole draft we we we we sent it to BBS and the rest is kind of History so we had yeah so okay eight reviewers I can tell you about that yeah that that that is crazy I've had three well that's a lot eight is I mean obviously that's my I haven't got anything out yet so these are all in process three is three felt like a lot eight is outrageous um well the good thing is that the more reviewers you have the kind of the more of a Arena it becomes where you can try to pitch them against each other in your responses right so you can and and we did have in in our reviews we did get you know reviews where One reviewer said cut this out this is irrelevant and the other reviewer said oh I find this bit really nice I'm really happy that it's there so right well okay so I mean this this paper I mean this I feel like when I read this paper I view it in concert with um The Mark of blanket trick so I I view those the these as kind of coupled uh critiques of active inference or predictive processing critiques I actually think is H it probably is a critique um I actually I read these as more like specul they're not speculative in a derogatory way but they're calling into question right they're saying okay let's let's get clearer on the terms that using so for example a distinction that you draw here is between a so-called marov blanket and a friston blanket precisely what what just in sort of layman's terms without getting into you know the granularities of Pearl and I'm going to ask who did the math in that

paper because I'm kind of amazed you know I looked at the list because sometimes you see there's like five philosophers and a mathematician and you know they've done all the formula formula and I was like oh no there's four philosophers here I'm like this is really cool that they've also done this um without like getting into the granularities of the variables what is the difference between a friston blanket and a mark of blanket like in short yeah okay so in very short terms without uh going into the details of you know Pear's work so so the main idea in graphical models is that sorry in basian graphical models I should say Invasion directed graphical models is that you have a number of variables that are related by some dependency probabilistic dependency relations and that's usually expressed by there being kind of arrows between the between the variables right so you have variables that influence other variables and then if a variable is dependent probabilistically dependent on another variable there be there will be an arrow that connects them pointing kind of the in the direction in which the influence go so if there's you know variable A up here that's modulating the behavior of variable B down here there will be an arrow pointing to to from from one variable to another and that will mean if there's an arrow connecting the two variables it will mean that the dependent variable is conditionally sorry I shouldn't say conditionally but it's dependent on the on the on the variable which is influencing it and so a mark of blanket in very simple terms is that if you have a graph structure where there are different variables connected and dependent to each other in different ways uh a mark of blanket identifies a set of variables which render some variable of interest a variable that in which the state of which we're interested in it renders that variable of interest conditionally independent from other variables in the model so basically it's a set of variables uh that render the variable of interest conditionally in the dependent from the rest of the model so what that means practically is that you have some variable that you're investigating and you're interested in learning everything you can about that variable that means that all you have to know about the rest of the system are the states of the variables in its Mark of blanket and that because that Mark of blanket is kind of shielding that variable from the rest of the system so if you know the states of the mark of blanket variables you know everything there is to to that is needed to make an inference about the variable of inference and that's basically all that Mark of blankets are right so Mark of blankets are basically uh a way to partition the variables in a directed graphical model probabilistic graphical model um partition them in a way where you will be able to effect efficiently and effectively do inference about some variable in that model um and so what's important about this is that the the the original formulation of a mark of blanket by Pearl that doesn't ascribe

any kind of special relationships there or special functions to the to the to the variables in the mark of blanket I mean I maybe yeah like all of all of the V all of the variables in the mark of blanket they can be connected in different ways of course probabilistic to the variable of interest but they don't you know none of them are kind of more important than the others so to speak um and they don't play you you cannot learn anything about the functions they play in the model just from their positioning in the in the mark of blank and now here's where the innovation of Carl friston comes in which is that Carl friston has partitioned the variables in the in the Mark of blanket into the into so first of all he partitions them in in two ways so he calls them the internal variable right uh and the external variables so so he assumes that there's the mark of blanket is kind of separating or shielding the internal variable which will used to be our variable of interest in pearls terms from the rest of the system which is external to the blanket so to speak and now the mark of the the second Innovation is that apart from talking about internal and external variables he also starts talking about uh ER he starts to talk about sorry and now I have to actually refresh my memory because this been a while since we wrote that paper I assum the sensory and active St yeah exactly I was just wondering what is exactly the term so not sensory act sensory States and active States right so there are the the states in the so variables in the the mark of blanket itself can take a kind of a sensory function or they can take an active function and the difference between these two is that some states some variables in the in the mark of blanket will take information from outside of the blanket and these will be the sensory States and there are some uh variables in the mark of blanket can influence what's happening outside of the blanket so they have kind of these istic arrows in the model coming out from from the from the blanket and influencing external States and these will be active States MH uh and this is this this is it seems at first as a kind of a innocuous you know distinction yeah sure so so what you're telling me you could say so you what you just told me is that like what Carl noticed is that some variables in the blanket are influenced by external variable and some variables in the blanket are influencing the external variables right and that would be all fine good if uh it wasn't that this distinct this partitioning into active and sensory and internal external wasn't pay playing an important role in in active inference in that it is taken to delineate a minimal active inference agent and this is kind of the leap from from Del from from uh calculating or delineating a blanket and going into it giving it an interpret interpretation of an agent that has some internal States and is interacting with the external environment via its perceptual and active states that that move from one to another is actually a huge leap because in Pearl's original formulation of Mark of blankets you can specify a mark

of blanket for any variable in your probabilistic directed a cyclical graph so as long you every variable in the model will have its own Mark of blanket and these will be partially overlapping of course in terms of the variables that make up the blanket but what's important is that no variable is kind of privileged in relationship to the other variables in the model in terms of uh it's it's how it's treated so so every variable has you know a mark of blanket is just a way for Pearl it's a formal tool for learning something about the dependencies in the model nothing more and now and doing inference of course over the states of Interest so it's kind of just a tool that's deployed in a purely pragmatic way and in fistance interpretation it becomes something more because suddenly there's this whole layer interpretive layer that's added to the mark of blanket formalism which implies that you know variables of Interest are somehow and Mar and blankets are constitutive of agents and it seems to and then implies that you know there are these agents and and the way that friston uh often kind of formulates his models there's an agent that's interacting with the environment and so the the the the interpretation of the mark of blanket under the free energy principle and in free energy models doesn't seem to be uh doesn't seem to be uh sorry seems to privilege a certain you know certain way of looking at what the model is supposed to be doing so it's kind of a built-in assumption that we're modeling an agent and this agent is doing something right in our case minimizing free energy right yeah yeah yeah so yeah this is this is interesting this is a conversation that's come up a lot in this podcast I wonder whether as a result of papers like yours um as well as a mark of blanket trick as well as Mel andrews's paper vanz other I wonder whether there's been a general attitude shift towards a well at least a recognition of this issue so I I've read the the the response as well and the commentary to the responses and you draw a distinction I think this is interesting between literalism realism and let me just check I don't instrumentalism realism and literalism and I like there's a graph I I don't know maybe it's just because I like graphs there's a nice graph where you sort of put everyone on the Spectrum um indeed where what can you in brief I mean I'm sure people okay so like instrumentalism I'm sure in part you know in short we can say okay the of blanket is a tool for us scientists uh but it's not actually making any claim to ontologically say what's going on in the world and then realism or sorry literalism would be well no the mark of blanket literally is a partition in the physical world what is the addition of realism there so that's a kind of the the Spectrum and then you got realism in the Middle where how does realism carve up that space well so I guess uh the the way I would think about realism is that um that uh that they they identify so so the realism would be kind of a position intermediate position where where you say well there are cases where in which

blankets do align with or are can be used to identify real features of the world right so so are are set to identify boundaries in the world uh the difference to literalism is that in this interpretation a mark of blanket is still a tool kind of a a a way to to use a way to capture a real difference in the in the world right uh but you cannot you someone who's a realist at least in my understanding and maybe my coauthors would have a different understanding also time has passed so we opinions but my understanding here would be that literalism is this type of claim that at some point appeared in the literature that you know an organism has a mark of blanket and I think that if you say an organism has a mark of blanket you're committing a kind of category mistake yeah because Mark of blankets are properties of are formal properties of models MH uh and not of organisms and so a more moderate position from that is to say well there's a boundary there's a real boundary in the world and we can delineate that boundary using represent that boundary using a mark of blanket right yeah and that under some conditions and like this is really important because I think we got a lot of push back for that so we we don't we never kind of argued against The View that you can create a model system where under some conditions there will be boundaries in this emergent boundaries in the behavior of the system such that you can model these boundaries with Mark of blankets effic effectively and I think this would be a kind of a realist position what we kind of argued against is that some people have claimed in the literature before we wrote that paper that the all this talk about agents uh having boundaries separating them from the environment and having internal external States and so on all this talk Falls naturally from the math from the math of probabilistic inference and variational free energy minimization and what we wanted to show is that that's not actually true if it falls from something it falls from some extra assumptions that you bring into the picture it wasn't you know it's it's a it's a it's a result of assumptions that you build into your models sometimes and that's fine if you if you're transparent about that there's no problem with that but it's not that you know you just have a a system where you can delineate a probabilistic graph you you can partition the variables and suddenly you show that oh my God there's an agent in there right that's that's not it just doesn't follow in this straight way and so I think this would be the the clear difference here between instrument alism also in literalism where instrumentalism is you know just being as non-committal as possible saying we're just doing modeling we're not claiming that what we're modeling needs to correspond in any uh in any truth preserving way to what's out and there in the world then there's realism where we say we are doing modeling but we do claim that we're tracking something real in the environment using this formalism and then literalism is saying the model is what is being

modeled right the medium is the message so to so to speak so so the blanket Mark of blankets are something that organisms have that are boundaries in the world that we can talk about boundaries in the world as Mark of blankets being out there and that's just nonsense in my opinion at least so okay okay so yeah so this is the point right is it that's the last point that that last point I think is the really critical one uh why is it nonsense that a coupled agent Arena relationship has a mark of blanket like what what is the part so there's one argument which is that it's a tool so it started off being a tool and it would be as you say it's a category mistake to confuse the map with the territory in this case so to confuse the tool with the actual on the thing that exists now that itself is not an argument against the thing that exists the thing exists being isomorphic to the map the map being the of course it's just saying it's just saying hold on right be careful right it's just precaution now it takes a further step to go well actually I think that it's nonsense that the map is the territory and I think this goes a little bit to what um V Vicente rager and his colleague spoke about that there are certain systems at least in his perception that it's not just the case that the tools that we bring to bear on these systems um you have to be precautionary when you apply them and say that there's something onto logic happening he's saying that the tool is insufficient right like whatever the tool is doing it's in no way M there's no way we can even it doesn't even make sense to say that something like that is going on in the real world and he has the example of a candle flame right and this idea with the candle flame is that the flickering of it implies that this Mark of blanket is kind of Shifting in and out of existence there are further arguments I mean they won't really pose as arguments but Andy Clark's things about metamorphosing Mark of blanket your own Mark of blanket super interesting papers Mike Le's work on butterflies and cater like questions of identity very philosophical questions but I'm kind of curious about how you go from saying okay it's an instrumental tool now I could be agnostic about how much the map should be conflated with the territory but it's a tool and it should be treated as a tool and we have a different name for what's actually going on in reality and it might be the same thing but let's use a different name so these are just very scientific questions mhm it's a why is it why did you take the further step to say actually no it's not just that it's a tool and we shouldn't apply it for scientific reasons but actually the to it it's literally not the territory uh yeah so um okay so I I think there are two things I would like to say about that so so first would be like an intuition pump right so so me saying that I have a that I have a mark of blanket would be something as ridiculous as me saying that I have number two what does it mean that I have number two right so so if if you like if I tell you I have number two you're going to say in what way in what sense do you have

number two like do do you mean that you you possess the concept of number two or that you can you know that you can calculate with number two or you have something that is uh that is shaped like numbered like the Arabic numeral or I have two items that essentially you know that's like what do you mean by that and I think that so so the the intuition here would be saying that you have a mark of blanket is very similar to to claiming that that you have number two like it's you it cannot be set without further qualification so in this sense it's nonsense because you know you have to you have to you have to say something more you can't just stop there so that's that's one point and now the the second point is I guess yeah so so so the the the the mark of blanket trick is a is a very good paper and I must admit that I personally I am not uh equipped uh in terms of my knowledge of physics and uh modeling to to to to to assess the that the arguments they said that they use there to to claim that the tool is completely useless for the task that U that it's supposed to do in the active inference framework sorry but um I think that my my and I think we would all agree here speaking for my author my co-authors I think that um what we would say about this something like this so what we would say here is well I I we we have demonstrated that Mark of blankets as such are are instruments formal instruments that can be that do not privilege uh between any element of the of a graphical probabilistic model right so every element every variable in a probabilistic directed graphical model can be described as having its Mark of blanket or a mark of blanket is better to say a mark of blanket of that variable can be identified yeah and so if this uh if this model does not if this tool does not disambiguate between or does not privilege any of the variables in the model that means that uh when you're trying to when you're moving in the realist Direction and when you're trying to find when you're claiming that there's a boundary in the environment that will model and capture using this notion it means that you're building in some assumptions which will help you privilege certain variables over other variables in in your model and that's as long as as I said before as long as you're staying uh you know honest about that there's nothing wrong with that scientists do that all the time basically building any model requires you to say like Hey we're going to build in some assumptions and that's fine and now I guess what I find troubling and and and where the problematic jump would be is to say okay so if you're if you're um if you're honest about the fact that you're building a representation of a certain object in the world right and you're honest about the assumptions so the idealizations and the kind of you know falsities and the simplifications that you've built into that model if you're honest about that it's near impossible to make the jump into literalism and claiming that you know like you you have to be so in order to be a real a good realist let's

call them has to be honest about the assumptions that go into their model and has to be honest about the fact that you know all models lie in a way in some way and in order to jump from there to to literalism you have to become a kind of a dishonest realist so you have to you to abandon you know hide all the assumptions in order to be able to do because because otherwise it's just ER otherwise I think at least to my mind it's it's just this kind of epistemic jump is impossible to make if you're taking in all the assumptions and all the idealizations and all the simplifications if you're stating them explicitly in the premises and you're clear that they are simplifications and idealizations then it's just basically impossible to make this final move to claim that the map is the territory because you just admitted that it's a map right so so you you kind of um yeah I so okay so yeah I understand that in terms of a strict literalism that the world is always too complicated in some sense it always udes us uh and so when we're modeling it we always have to build in our assumptions but it still seems to me that there's a difference between assumptions that lie and assumptions that simplify right so I can I can take a very coar grained uh model of reality which never nevertheless has a faithful reliable mapping to that reality like a map right and and so maybe I end up having a kind um teleonomic analysis of my map that it serves me right like my GPS serves me it isn't the world but it serves me now that seems very different from a map a mapping which lies right a map which says that I'm in Paris when I'm in London now that's just to one that's totally useless and two like it's literally wrong it's not just that it's a Well it's not just that it's a compression it's a compression which in its ISO you know which in its mapping is wrong right I'm not saying that the mapping I'm not saying that of course a compression is right of course there there's not a strict isomorphic mapping between the map and the territory nevertheless I still think I I'm curious about whether you could whether you might be open to the idea let's say that nonetheless we build in our assumptions they're simplifying so we can't adopt a strict literalism nevertheless there is a way in which we could build out of that model that we end up having given the assumptions that we've had and we inflate it all and we end up with a picture of reality like an actual true faithful picture of reality and that follows basian mechanics um okay so well it's worth I have no invested uh usion in this game right like I'm I'm just asking questions so just on the table I just think this stuff is interesting yeah no no this is this is this is a perfectly fine question and I think it's it's uh great that you're asking it it's just that I'm trying to uh figure out you know so so so your question basically is that um well so so I think we have to we have to ask ourselves what is the end goal here maybe this is this is what we could start with because so literalism would be a view that kind of like in the B has famous short

story about the Empire and the map of the Empire literalism would be a view where the the you know you blow up the models you you you you kind of build that build up build up build up build up you describe more and more phenomena using those models and suddenly it turns out that you've it's the world the model it's the world exactly so so there would be a one-to-one correspondence between the model and the world and I think that first of all that's not really feasible well it's not a model exactly it stops being a model and then the second thing is that um it's it stops having any use so to speak right uh so so it becomes useless so as as soon as the map of the Empire spans the whole empire it's it cannot perform its function as a map do you think there are people the active inference Community who think that liter you know adopt such a radical literalism no no no but but but I I'm sure that you know the literature has exploded expanded so much that I'm pretty sure pretty sure that any position you can imagine in that literature has been taken by someone so sure but uh but when it comes to the but but people do talk about you know organisms having Mark of blankets and I find this very puzzling so this is just what we were attacking this as we said earlier category mistake now the the interesting question here the there's a second possibility here which I think you're pointing to which is that we're building we're starting with a model of the world which you know is not not an exact replica of the world it's a it's a model and then we kind of expand this model and we end up with something that is not an exact replica of the world because it shouldn't be but it's is a complete system um that corresponds in interesting ways or in relevant ways or some truth preserving ways to how things are in the world so we kind of get this this this this model that tracks things and and in a realistic way and I think that's uh I'm not close to this possibility I just don't entirely see right now how you could get there so to speak with the with the mark of blankets and active inference at this point in time so one of my worries that we don't mention in the paper I think is I don't remember now exactly but I don't think it's in the paper one of my worries is that um you can have two types of explanations right so you can have this kind of horizontal view of explanation where you're explaining you're covering a huge ground and you're bringing a lot of phenomena in under one kind of shallow umbrella where you're creating this kind of uh spreading structure that connects many many often very dist disci by using a kind of a lingua franca of of some formalism and this seems to be the project of of active inference right now so there's more and more fields that are kind of brought Under the Umbrella of free energy more and more phenomena are described as versions or as you know um of ways of doing active inference and that's very interesting it's an interesting unificatory project where you're trying to unify

many different branches and many different domains of science but there's another model of explanation in science which is not about unification but is about vertical integration right so you're kind of trying to study one phenomenon on different scales and and you're doing it in a very kind of narrow window uh narrow narrow kind of uh window of focus so you're really trying to understand how one mechanism is realized on different levels of organization so neuronal spiking right for example you can study the ion channels and then you you can study protein folding and all these kind of different aspects of how a neuron operates and you're just interested in one in one piece of the brain you're interested in how a single neuron responds to stimulate but you can study that on ever more fine grain levels of description and in the end hopefully you have this kind of vertical integration where you understand how the brain works starting from a very you know s a subcellular level onto onto the level of the whole brain or at least an area of the brain or a network of neurons uh and I'm not entirely sure that these two models of explanation are compatible so I think that active inference and the free energy principle is doing an interesting interesting work bringing in different domains of inquiry into a common framework in which they can start to to interact so suddenly you know it's it used to be the case that people doing reinforcement learning were very far from people doing I don't know something like uh phology yeah for example yeah uh or or doing perceptual modeling and suddenly there's a possibility that these two people these two groups of people will interact and will be you know talking to each other and that's extremely valuable however I'm not entirely sure that the free energy principle and the active inference stuffff is very good at providing this kind of vertical integration precisely because as you're trying to describe everything using the same tool it might turn out that on different levels of description there are better suited more accurate ways of modeling things that don't conform to the assumptions of the free energy principle or the assumptions of the active inference framework in general whatever you want to call it so there might be kind of in different domains of inquiry kind of these vertical domains of you know where people are trying to study a particular phenomenon at a ever finer level of detail there might be domain specific or level specific ways of modeling certain certain phenomena or processes which are better suited for that purpose than the free energy formulation using Mark of blankets and then the problem becomes kind of that well should we abandon that in favor of this overarching you know structure that connects everything I think we can be a kind of pluralist about that so we can we can think of the free energy principle or the active inference framework as this uh highlevel abstract unifying layer while not uh diminishing the work that's being done in very particular uh areas and the reason

sorry just I lost track a little bit because I'm getting tired yeah soon but the reason why I'm not sure that active inference can do this vertical integration is because precisely it's trying to redescribe every phenomenon on every level of finess of grain as doing the same thing yeah and that's actually a problem right because if you find out that a single neuron is doing the same thing that a that I don't know an ion channel is doing you might be in a you might have a problem here and I think that's that that that's maybe my my general qualm here yeah so I'm getting tired as well so so we will wrap up imminently um but there just several things that come to mind I mean one is that I think it's probably not the case at least in most recent formulations that active inference would say that the neuron is doing the same thing as a planet it's just that the neuron looks like it's doing the same thing as a planet or can be described as if it's doing the same thing as a planet and then I guess all of these concerns get deflated in some ways which is which I mean look given my again you have been in this game a lot longer than I have but even my first encounter of active influence is very much being on this looks as if line so I think we're not really going to have any problems with that and even in your commentary your response to the 35 people who got in touch uh you very much say you know in response to inz and Maxwell and Thomas vanz like we don't really disagree like it's instrumentalism deflates all of this I guess the thing is right like you said something really interesting in there and I'm wondering whether it's a like whether you meant to say it which is one is that there may be tools which are better at modeling right so I actually hear this a lot from RL people where I work at Royal Holloway which is active influence is an as if it's very good it's very interesting it's cool that you can get from basic physics to uh a as if physics of life like it's cool it's interesting however uh we have to do model comparison and if active inference models uh better than RL then we'll use it but if RL is better than active influence we'll use that now that's that's one thing and and like I think everyone in the scientific world is interested in that and and should probably advocate for that because why not right like it has better Downstream uh effects when it comes to whatever therapeutic goals but you also said something that is that sometimes those models those RL models uh again someone can go back and check the tape but it was something akin to and you can check you can correct me uh it they may actually break the assumptions of active inference now that seems a different that's true well I guess my question here is is it a different problem because so if I break the Assumption of a theory is it does it turn out that that theory is wrong so if for example um active inference rests on I don't know whatever it rests on like okay it rests on I mean maybe it does again I think another issue here is are we talking about the free energy principle are we

talking about active inference as predictive processing active inference as P active so I've actually been thinking this in my own case uh Act of inference is a very loose term right when I write it I say it's a corollary of the free energy principle because KL says it's a COR and I know what that means but and then they say active inference is a process Theory I mean that 2017 FR and paper is active inference as a process Theory now the free energy principle is not a process Theory so active inferences is kind of this weird Middle Ground of broaching Bridging the free energy principle predict to processing and pomdp schema but you know that said I guess the question is right if I take one of these deep assumptions of active inference for example we are doing uh there's a KL Divergence so rather than doing Monte Carlo sampling we use the K Divergence if I break that do I in turn break active inference and and how deep does that rot go because right like in my estimation I've said this on this podcast Cole said it to me the free energy principle is quite shallow like it's very deep and shallow in an interesting way it's deflation it doesn't really tell you anything that exists and then you build out of it I'm wondering whether you say Okay RL breaks a core assumption I don't know what that would look like but let's say it breaks a core assumption um how deep does that rot go okay so what I was thinking about specifically is that if you want to be a real realist about Mark of blankets you have to bring some assumptions on board right so like you you it's not like it all falls out of using variational base you have to assume some extra stuff and so in the past one of these assumptions was ergodicity I was gonna say ergodicity yeah and so ergodicity is a good example where if you have a model in some domain at there was a time I don't know what it is now because you know I've kind of work on other stuff now so I have been paying less attention to the development in the literature but I know that people have been moving away from the ergodicity Assumption and the in the inference literature but there's a there's but ergodicity is a good example because there was a time uh where ergodicity was kind of baked in to what people thought active inference and the free energy principle are supposed to be and then if you break that assumption so if you don't have that assumption in your model then it would seem that you know you're you're you're breaking some you're incompatible on a very important assumption with with what's happening in the what was happening at the time in the active inference and so the question would be then like does that completely invalidate your model or whether that completely invalidates the free energy principle and I think I would be rather reconciliatory here by saying that you know you don't expect all the models to have all the same assumptions because models are representations of reality so what you probably want to do is like there's some phenomenon that you're trying to describe and the more

complex the phenomenon like life itself the more angles you want to look at it from so you probably want to have this plurality of models and I think that's that's all fine right uh the other thing is uh and then if if you're a pluralist in this way I guess the problem you might encounter is that why would you need a single principle like the free energy principle right yes uh in that kind of raises different questions and and again if if you're a pluralist if you're a kind of a perspectival list where you think that there are multiple perspectives to look at single phenomenon that are can be all equally informative depending on your explanatory interests or pragmatic interests then I would say Well it starts to become it starts to be difficult to see why you would want to claim that there's this overarching principle yeah unless you're looking at this Principle as just a means as I said before of building this super structure right which is just you know as if doing free energy minimization story which is supposed to create these Bridges between very distant disciplines or domains bringing them into together into into conversation which is pragmatically very valuable let's agree on that but it's it's it might not be the be all end all of you know science no no I mean I mean yeah there's an interesting kind of meta sociological point about why people want to Gran unifying theory of anything I mean it's it's definitely satisfying um in some primordial sense but beyond that it may be wise to rest among um plurality as you say and and there's also the you know the argument from the pessimistic meta induction which is that uh every Theory up till now so we have certain set of scientific theories we think are uh best right now but every theory that preceded them in each of the fields has been discarded as false or as you know missing some important aspects so why should we assume that our current theories are true if history of science is is you know layered with with with husks of abandoned theories and one more thing that I just wanted to mention you earlier said about the incompatibility of different theories in science and I immediately thought about the incompatibility of quantum mechanics and and Newtonian mechanics and what's interesting about that case is that even though we don't really know how to make the two compatible uh we haven't abandoned either right so there there's like there's a possibility that you will have on different levels of description of reality you will have different models with different assumptions describing different phenomena and that's actually okay maybe that's just how it is yeah I I'd like to get a philosopher philosopher science on here because I think um this isn't really well this isn't my era of expertise but I I think there are slight differences between the models that employed by the free energy principle and the models that Newton employed and I think one of them is that when I write down uh f equals ma right the of course I make the Assumption of you know and I do that school and I they say okay what's

the acceleration of the car and it's got this got this Mass I then make the Assumption of uh win resistance and um the lack of wi resistance Tire lack of friction and everything like but it seems to me that in principle Newton could have recognize those assumptions and built them into his model and he gets a closer approximation of the truth the question therefore and we'll leave it here because we been going for three hours and a bit is whether the free energy principle is like Newtonian mechanics such that it can act as the base a true ontological base because I think if you ask Newton he would say literally $f = ma$ right like it's not just I say it and it looks as if things are doing $f = ma$ it's like if I know like it's not just if I know M and I know a going to know F it's like that's a law of the universe inscribed in the universe now I'm not a philosopher of science I'm also not a physicist so I might be wrong here but it seems to me that that therefore acts as a base on which you can build in friction you can build in air resistance and so on and you get a closer approximation on the truth and that doesn't make his theory false right because it idealizes it just makes it a base right and it just means it's pragmatically useful but it doesn't make it false the real question here therefore and I'm not going to answer it today because I'm out of caffeine and I'm out of energy is whether the free and liy principle is that base as well and I guess uh time will tell on that front but you know I mean ultimately epistemic humility I do think is the way right I mean I think I think if we can talk about one recurrent thing that's been very useful in the history of science is epistemic humility plurality listening to the other side and uh so that's why you know I very much appreciate you coming on the show um because well it's not a lion's den I mean I've had viente on I've had all sorts I've had people who don't I think most people are partial to active inference but it's not really a strict active inference fan club um but no I mean I really do appreciate the openness and the expertise and the dantian curiosity now it's um it's been amazing I mean three hours and five minutes have flown by thanks yeah it's sorry that I wore you out uh so much you should you should be apologizing if you didn't wear me out um uh yeah you just to close up you asked me at some point who wrote the maths in the mark of blanket paper so it was first and foremost Manuel baltieri because he has background in machine learning so he knows a lot of this stuff and then I was helping him I think that was the that would be the correct way to nice okay yeah always I'm always I'm always super impressed I mean it's just it's kind of a project that I'm trying to undertake myself as well orbe it having that and then simultaneously battling the kind of fears like no this is just an instrumental tool get back to Consciousness get back to Consciousness like that's for not like you know at least that exists and so on and there's an interesting uh sort of pulling and you

know throwing in different directions and I guess active inference does that it Bridges as you say these multiple domains but I'm very glad personally people might disagree with me I'm very glad that philosophers have brought to bear their expertise um I think as you know in the modern world we probably are a bit disenchanted by philosophy but actually I think with the prominence of active inference it's kind of shown its strength and power to say hold on a second think about your presuppositions think about your axioms and I'm personally a really big fan that of the fact that phenomenology and Consciousness has also kind of returned I mean it's not that you know it returned in 1995 or you know you can go Thomas Nagel but like this feels like a real Renaissance of this question um so you know Long Live philosophy I think is the is the final note but yeah Chris amazing I I think this much well I think this is the longest podcast um no no no no well Cole went no way Cole went for about three hours Julian kinee went for about three hours so you're you're in good company okay thank you really I this was really really fun I really enjoyed it so thank you so much uh all right thank you so much for your time again sorry for wearing you out sorry if I was maybe a little bit incoherent at times which sometimes happens when you're going for so long uh please send me the draft of your paper you have my email so I would love to read that that's very kind and also Al uh before you leave and before we end the show please let people know where they can find you what you've got coming up you'll work with ax or like anything that you want to say in terms of self right uh I know this horrible everyone hates this bit but no no it's yeah I I get it that's how it works um so my website is Chris k r Ys dolega doxyz uh that's where you can kind of track my Publications and stuff and if you look for chrise Lega spelled in the same way on Twitter or blue sky and Mastadon I think that will also pop up um and uh yeah together with Axel we're organizing a symposium at this year's assc in Tokyo oh nice so that's something that uh if you're in Tokyo at the time and visiting for the assc you can come to it's on unconscious affect and finally I will be also giving a talk which will be online uh in something called uh the Tokyo analytic philosophy Forum uh let me just check if I remember the date uh I will make sure all of this is in the uh Tokyo forum for analytic philosophy it's it's uh series of talks that's hosted at the Tokyo University and I will be speaking on on the 19th of July fantas actually about something that we touched on briefly here so about effective modes of presentation and perceptual experience so yeah if you're interested in that you can join that will be that will be online will be going in this will be going in the description assc will be going in the description I'll put the website in that as well okay cool cool we've done three

hours and 10 minutes okay amazing thank you so much thanks a lot have a good evening and take care